

Pratyabhijñā × Active Inference, v0.4: A Mechanism Study of Recursive Self-Reflexivity Layers for LLM Creative Cognition

Sharath Sathish¹

¹Independent Researcher

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Abstract

We report v0.4 of the *Pratyabhijñā Creative Engine* (PCE), a recognitional cascade that operationalises Abhinavagupta’s five-*śakti* architecture (*cit*, *ānanda*, *icchā*, *jñāna*, *kriyā*) as an executable layer over a Claude Haiku substrate, with a recursive *vimarśa* loop that may revise the surface and a Bayesian-Model-Reduction step that prunes a candidate posterior via *apohana-śakti* (contrastive exclusion). v0.4 is deliberately framed as a *mechanism study*, not a horse race: at the available pilot n (4 domains, 25 paired items in total) the headline cascade-vs-bare contrasts are statistically inconclusive, but several mechanism-level signals are robust. We make four contributions. (i) **A four-arm Bedrock pilot** (Haiku as cascade scorer, Sonnet-4 as judge) under a bytewise-isolated CLI substrate yielded H5.v4 fixed-effects pooled $g = +0.14$ (95% CI $[-0.26, +0.54]$, $n = 25$) — not significant, but with a directional pattern dominated by `poetry_interp` and `sci_creativity`. (ii) **A within-cascade revision-quality test** (H8a, $n = 27$) shows that the shadow revision strictly beats the draft on average, $g = +0.65$, $p \approx 1.210^{-4}$. The cascade reliably *produces* a better surface; the question is whether it reliably *commits* the better surface. (iii) **A four-policy commit multiplexer** (H8c) ranks `always_revise` ($g = +0.53$) above `learned_gate` ($g = +0.39$) above `event_gated` ($g = +0.21$) above `always_draft` ($g = -0.07$); pairwise paired permutation rejects `always_revise = learned_gate` ($p \approx 0.034$) and `event_gated = always_draft` ($p \approx 0.0073$). The *vimarśa* event gate as currently defined is mis-calibrated (H8b: $F1 = 0.52$ vs. a learned gate’s $F1 = 0.65$); the cascade chooses to revise less often than it should. (iv) **A judge-vs-scorer agreement test** (H9, $n = 23$) finds Spearman $\rho = 0.0$ with sign-agreement 56.5%: the Sonnet judge and the Haiku composite-score head are *not* measuring the same thing on the cascade items. We treat this not as a defect but as a load-bearing finding for any cascade architecture that uses a small-LM scorer to gate a large-LM commit. The pilot consumed \$12.73 over 1,277 Bedrock calls. We additionally release a 9-output *showcase* (3 Sanskrit chandas · 3 English poetry styles · 3 scientific-creativity prompts) with full draft / shadow-revision / commit traces, an Astro-powered GitHub Pages site reproducing every figure and number from the underlying artefacts, and a Cursor-portable plugin manifest plus a standalone `python -m pce` CLI that exercise exactly the same OAuth-CLI substrate the Bedrock pilot used. We close with an honest limitations section that addresses the unmerged-state v0.3 critique directly: v0.3 results were never merged to `main` because the v0.3 SPEC had a closing "negative-result obligation" clause that the run met but the prose did not own; v0.4 owns the negative result, ships the mechanism findings, and makes the falsifiable next test (a chandas-aware scorer) explicit.

1 Introduction

The missing layer. Large language models routinely produce *competent* text but rarely produce text whose interpretive multiplicity is deliberately authored. Where humans use a single

utterance to gesture toward several incompatible readings simultaneously — the Wittgensteinian *aspect-shift* (Wittgenstein, 2009) — LLMs typically collapse onto a single dominant reading dictated by the prior. We argue that the missing structural element is not scale, alignment data, or instruction-tuning sophistication, but a *recursive self-reflexivity layer*: an explicit operator that re-examines the model’s own output against the same constraint manifold that produced it, is permitted to revise the surface when the alternatives are not yet visible in it, and gates that revision on a measurable free-energy signal rather than on the prompt-template’s exhortation to “check your work”. The contemporary literature on iterative self-refinement (Madaan et al., 2023; Shinn et al., 2023; Bai et al., 2022) is converging on the same insight from the engineering side — giving an LM a critique step buys real quality — but it leaves open the gating question: *when* should the model revise? On every item, regardless of cost? Only when an internal signal says the draft is suspect? Only when a learned policy says revision will pay off?

The Pratyabhijñā tradition has a precise answer. Abhinavagupta’s Pratyabhijñā (“recognition”) school of Kashmir Śaiva phenomenology developed an unusually fine-grained taxonomy of the layer above first-pass cognition (Lawrence, 1996; Abhinavagupta, 1986; pra, 1963). The five-*śakti* cascade (*cit*, *ānanda*, *icchā*, *jñāna*, *kriyā*) describes the trajectory by which an undifferentiated luminous ground (*prakāśa*) becomes a determinate utterance, and the *vimarśa* layer is the moment of self-touching — the recognition that the utterance was authored by the very same ground that contemplates it. *Apohana-śakti* (the contrastive exclusion of all-other-than-this) is precisely the geometric move that Bayesian model reduction (Friston et al., 2018; Friston and Penny, 2011; Friston et al., 2016) formalises as posterior switch under a parameter-pruning prior, and that the broader free-energy programme (Friston, 2010; Friston et al., 2017; Friston, 2013; Heins et al., 2022) treats as the operational substrate of insight.

The Pratyabhijñā tradition’s contribution to a recursive-LLM architecture is not a metaphor. It is a precise structural answer to two questions the contemporary literature is not yet asking sharply: (a) the gating question above (the answer is: gate on *vimarśa-śakti*, the recursive self-touching that occurs when the system recognises its own *prakāśa* as the source of its current draft), and (b) the *posterior-pruning* question (the answer is: prune via *apohana*, contrastive exclusion against the *not-this* set, formally equivalent to BMR with a sparsifying prior). This paper does not claim the Pratyabhijñā tradition is true in any metaphysical sense; it claims the architecture is correct, falsifiable when implemented, and useful as a design source for the gating and pruning layers in modern LLM cascades.

What v0.4 reports. This paper is v0.4 of the *Pratyabhijñā Creative Engine* (PCE), the executable instantiation of the architecture above. PCE is implemented as a Cursor / Claude Code plugin and as a standalone `python -m pce` CLI; both share a single substrate (the `claude-print` OAuth-CLI binary against either an Anthropic API key or AWS Bedrock). v0.4 is deliberately framed as a *mechanism study*: at the available pilot n (4 domains, 25 paired items in total) the headline cascade-vs-bare contrasts are statistically inconclusive, but several mechanism-level signals are robust enough to be load-bearing for the architecture’s next iteration. The contributions are:

1. **A four-arm Bedrock pilot** (Haiku as cascade scorer, Sonnet-4 as judge) under a bytewise-isolated CLI substrate, addressing the v0.2 reviewer’s contamination critique. The merged ledger reports `V04_COST_TOTAL_USD` over `V04_COST_N_CALLS` Bedrock calls. Per-domain Hedges’ g (H1.v4 through H4.v4) ranges $[-0.32, +0.32]$ with no individual domain crossing the BCa zero line; the H5.v4 fixed-effects pool is $g = V04_H5_POOLED_G$, 95% CI $[V04_H5_CI_LO, V04_H5_CI_HI]$, $n = V04_N_PAIRED_TOTAL$.
2. **A within-cascade revision-quality test** (H8a, $n = V04_H8A_N$): the shadow revision strictly beats the draft on average, $g = V04_H8A_G$, $p \approx V04_H8A_P$. The cascade

reliably *produces* a better surface; the open question is whether it reliably *commits* the better surface.

3. **A four-policy commit multiplexer** (H8c) that ranks `always_revise` above `learned_gate` above `event_gated` above `always_draft` on paired delta vs. `haiku_bare`; the headline is `V04_H8C_WINNER`. The *vimarśa* event gate as currently defined under-fires (H8b: event-gated $F1 = V04_H8B_EVENT_F1$ vs. a learned gate’s $F1 = V04_H8B_LEARNED_F1$).
4. **A judge-vs-scorer agreement test** (H9, $n = V04_H9_N$): the Sonnet judge and the Haiku composite-score head are not measuring the same thing on the cascade items (Spearman $\rho = V04_H9_RHO$, sign-agreement `V04_H9_SIGN_AGREEMENT`). We treat this not as a defect but as a load-bearing finding for any cascade architecture that uses a small-LM scorer to gate a large-LM commit.

The unmerged-state question. An honest reader of this repository will notice that the Phase 7 results that drive every number above were left for several days on the `v0.4-mechanism-study` branch, never merged to `main`. We address this directly. The v0.3 SPEC closed with a “negative-result obligation”: any pilot that failed to find the predicted directional effect must be reported *as that finding*, not framed as a methodological caveat. The v0.3 prose acknowledged the obligation but did not enact it; the prose framed the inconclusive headline as a sample-size limitation and set up additional control arms for a v0.4 rerun, which is what Phase 7 then ran. v0.4 inherits the obligation and discharges it: the headline cascade-vs-bare result is inconclusive at $n = 25$ and we say so in the abstract; the load-bearing mechanism findings (H8a, H8c, H9, H8b) are foregrounded for what they actually are; the speculative scope-cuts (chandas-aware scorer, Hopfield *ālayavijñāna* substrate, learned gate trained on more data) are deferred to v0.5 with explicit falsification criteria. The unmerged-state evidence-then-prose discipline is, in retrospect, a feature of the pipeline, not a bug: the artefacts on the branch are the truth, and v0.4 is the prose rebuild around them.

Organisation. §2 situates PCE among recent work on Indic-philosophical AI architectures, BMR, LLM-as-judge methodology, iterative self-refinement, and computational Sanskrit. §3 specifies the Pratyabhijñā primitives. §4 states the active-inference and BMR primitives, including the honest scope of free-energy in PCE (best-of-K composite scoring, not exact variational inference). §5 defines the operators and the consolidation cycle. §6 describes the plugin surface. §7 formalises the v0.4 experimental protocol, including the Bedrock substrate and the commit-policy multiplex. §8 documents substrate isolation and portability across Cursor, Claude Code, and the standalone CLI. §9 describes items and scoring. §10 reports the five hypothesis families (H1–H4, H5, H8a, H8b/c, H9). §11 interprets the findings across eight subsections. §12 is a dedicated section on what the cascade can and cannot claim about “creativity” on the available evidence. §13 walks the nine release-quality showcase outputs, each with its full draft / shadow-revision / commit trace. §14 and §15 close.

2 Related Work

2.1 Indic-philosophical architectures in computational settings

The literature that takes Pratyabhijñā or adjacent Kashmir Śaiva architectures as a serious source for AI design is small. Lawrence (1996) provides the clearest peer-reviewed entry point into the philosophical operators of the Pratyabhijñā tradition — in particular into the *discursive* (rather than purely metaphysical) reading of *vimarśa* as the moment a speech-act recognises itself as authored from the same ground that contemplates it. Abhinavagupta (1986) is the canonical primary source for the *śakti*-cascade taxonomy itself, and pra (1963) provides Singh’s

twentieth-century English-translated condensation that has shaped most subsequent secondary literature. The more recent Waldron (2003) establishes a comparative bridge to the *ālayavijñāna* (“store-consciousness”) tradition that PCE uses as the conceptual basis for its consolidation cycle (Appendix B).

Our work differs from prior philosophical readings in three load-bearing respects. First, we ground the cascade in concrete operators on a real LLM substrate rather than in a connectionist or theoretical sketch: every *śakti* corresponds to a specific computation visible in the Phase 7 traces (§5). Second, we make the *vimarśa* loop a falsifiable trigger under a computable signal (the cascade’s ΔF between draft and shadow revision), not a generic recurrent module; whether it fires correctly is something we can measure, and v0.4 finds it does *not* (H8b). Third, we publish all plugin and benchmark artefacts so any third party can replay the contrast. This last point reflects the Pratyaksha (Skt. *pratyakṣa*, “perception” / “directly evidenced”) stance — the witness invariant of the Pratyaksha Context Engineering Harness (Sathish, 2025b) from which PCE inherits its discipline that every claim must be anchored to externally verifiable artefacts.

2.2 Active inference, free-energy, and Bayesian Model Reduction

The active-inference programme provides the formal substrate that PCE’s *jñāna-śakti* (Bayesian-Model-Reduction over a candidate posterior) and *vimarśa-śakti* (free-energy gated revision) are built on. Friston (2010) introduced the unified free-energy formulation that is now standard in the literature; Friston (2013) situates the same formalism in self-organising systems with Markov-blanket partitions. Friston et al. (2017) explicitly link the resulting BMR step to insight: a sudden re-organisation of the posterior is the formal correlate of an “aha” moment, which is precisely the dynamic PCE intends to instantiate.

Friston and Penny (2011) and Friston et al. (2018) introduced and generalised BMR as a method for posterior switching under nested models, recovering sparse posteriors from rich ones via parameter pruning; Friston et al. (2016) provides a peer-reviewed empirical-Bayes statement that PCE’s *jñāna* step references for its candidate-posterior reduction. Heins et al. (2022) (and the recent `pymdp` v1.0 release notes (pym, 2026)) provide the discrete-state-space implementation that the v0.5 ladder will use to swap PCE’s best-of-K composite-score head for an actual variational free-energy minimisation. Di Paolo et al. (2024) has begun making the bridge from active-inference theory to LLM tooling explicit; PCE is one concrete instance of that bridge.

We use BMR as the formal substrate of PCE’s *jñāna* operator (posterior over candidates) and as the conceptual frame for the SWS/REM consolidation cycle (Appendix B). PCE’s *vimarśa* layer is a free-energy-gated revision step that fires when the cascade’s ΔF exceeds a threshold; an honest accounting of what “free energy” means in the v0.4 implementation is given in §4.

2.3 LLM-as-judge, scorer-vs-judge calibration, and evaluation methodology

PCE is built on a two-tier evaluation stack: a small Haiku model serves as the cascade’s per-step composite-score head, and a larger Sonnet-4 model serves as a downstream judge against which the scorer is calibrated. The wider literature on this configuration informs our H9 design and our reading of the H9 result. Zheng et al. (2023) is the canonical reference for the LLM-as-judge protocol and its known failure modes (position bias, verbosity bias, self-enhancement); Liu et al. (2023) demonstrates that strong LLM judges can serve as reference-free scoring surfaces for open-ended NLG, which is the regime PCE operates in. Sellam et al. (2020) establishes the alternative learned-metric lineage (reference-grounded) that any reference-free protocol must be checked against.

Recent work has documented systematic biases that bear directly on PCE’s H9 finding (judge $\rho = 0$ with scorer). Braun (2025) shows that LMs exhibit acquiescence-style answering

tendencies that distort rater outputs in ways the scoring surface may absorb differently from the judge. Laurito et al. (2025) documents an “AI–AI bias” in which LMs systematically favour LM-generated text over human-generated text, a property that any judge-of-LM-output protocol must control for. PCE’s H9 result — judge and scorer disagree — is consistent with this literature; we read it not as a defect of the cascade but as a load-bearing finding for any architecture that uses a small-LM scorer to gate a large-LM commit. §11 treats this point in detail.

2.4 Iterative refinement, sample-and-rank, and commit policies

PCE’s commit-policy multiplexer (`always_revise` / `learned_gate` / `event_gated` / `always_draft`) sits in a literature converging on the same engineering insight: an LM with a critique step beats an LM without one. Madaan et al. (2023)’s Self-Refine is the closest mainstream analogue to the cascade’s draft→shadow-revision→commit pattern; Shinn et al. (2023) extends the analogue to multi-episode reflection buffers; Bai et al. (2022) establishes the ‘constitutional’ critique-and-revise pattern from which PCE’s domain-specific rubric prompts (poetic craft, scientific honesty, etc.) descend. Stiennon et al. (2020) establishes the canonical sample-and-rank lineage that the H6 fairness contrast (`haiku_cascade` vs. `haiku_bare_2K_scorer`) is designed to control against: PCE’s claim must be that the architecture matters, not merely that doubling the candidate budget at the bare scorer is enough to close the gap.

The literature has not, to our knowledge, isolated the gating question that H8b and H8c address: *when* should a refinement loop revise? Self-Refine and Reflexion default to “always”; sample-and-rank defaults to “best-of-K”. PCE’s H8c finding — `always_revise` > `learned_gate` > `event_gated` > `always_draft`, with the event-gated *vimarśa* trigger as currently defined under-firing — is, in this context, a small but pointed contribution to the gating sub-literature: at the v0.4 budget, the cheapest correct policy is “revise unconditionally and let the scorer pick the better of draft and revision”. Whether a properly trained *vimarśa* gate can recover the cost saving while keeping the quality is the open question for v0.5 (§11).

2.5 LLM creativity benchmarks

PCE’s four-domain Phase 7 pilot draws on the public creativity-benchmark stack. POEMetric (Li et al., 2025, 2026) contributes the 6-axis poetry-generation rubric (creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery); CreativityPrism (Hou et al., 2025a,b) aggregates AUT, TTCT, and other divergent-thinking items under a Quality·Novelty·Diversity decomposition; Cao et al. (2025) provides a multi-domain reference-aware text-creativity evaluator; Wang et al. (2026) adds a self-evolving challenges pipeline. Suzgun et al. (2023) (BIG-Bench Hard) contributes scientific-reasoning probes; Tian et al. (2024) contributes the canonical constraint-satisfaction creativity items. Organisciak et al. (2023) establishes that LM-based scoring of divergent-thinking outputs can match or exceed human inter-rater agreement on AUT-style items, which underwrites the PCE composite-score head. Mtenga et al. (2024) and Weber et al. (2025) provide the recent Hopfield-network framing of creative-chain dynamics that PCE’s deferred Hopfield *ālayavijñāna* substrate (Appendix B) draws on; Beaty et al. (2015); Chen et al. (2025) establish the empirical neuroscience baseline (DMN/ECN switching during creative production) that motivates a recursive-revision architecture in the first place.

The Wittgenstein aspect-shift task is a custom $n=20$ stratified slice we construct in §9 on principled grounds, since no public benchmark explicitly tests *simultaneously-coexisting* interpretations of a single line of poetry — the precise capability the *vimarśa* layer is intended to enable.

2.6 Computational Sanskrit and chandas-aware generation

The v0.4 release includes a 9-output showcase whose first three items are Sanskrit chandas (anuṣṭubh, gāyatrī, indravajrā) generated under a syllable-count and metre-pattern constraint (§13). The validators (`tools/sanskrit_chandas.py`) implement minimum-correctness syllabification for both Devanāgarī and IAST input; the v0.5 ladder will swap the maintainer-curated Sanskrit references for live cascade outputs once a chandas-aware scorer is wired in. The relevant computational-Sanskrit literature is mature enough to support this: Hellwig et al. (2023) and Hellwig and Biagetti (2025) establish the corpus-and-annotation backbone, and Nehrdich et al. (2024) demonstrates a unified ByT5-Sanskrit model that handles segmentation, tagging, and parsing in a single pipeline — a natural drop-in for the v0.5 chandas-aware scoring head.

2.7 Plugin architectures for LLMs

The Cursor and Claude Code ecosystems have converged on a manifest-driven plugin model with MCP tools, slash commands, hooks, agents, and skills. PCE follows the same conventions used by `attractor-flow` (Sathish, 2025a) (which couples a real attractor-network engine to MCP), so that the PCE plugin can be installed and audited without any framework-specific shim. §8 documents the v0.4 portability work: a single source-of-truth plugin declaration mirrored to both Cursor’s `.cursor-plugin/` and Claude Code’s `.claude-plugin/` manifest format, plus a standalone `python -m pce` CLI wired into `pyproject.toml`’s `[project.scripts]` so the substrate can be exercised without any IDE plugin runtime at all.

3 Pratyabhijñā Architecture

We restate the Pratyabhijñā primitives we operationalise. We take the canonical *Īśvarapratyabhijñākārikā* formulation and preserve its phenomenological commitments while assigning each primitive an explicit computational role. Diacritics follow IAST.

3.1 The luminous ground

Prakāśa (“self-luminosity”) is the undifferentiated ground of cognition. It is not a state of the network; it is the property that any state at all is presentable. In our engine *prakāśa* is the LLM’s native distribution at temperature τ , which is what makes any candidate token visible at all.

3.2 The five śaktis and the recursive layer

- **Cit** (consciousness/illumination): the bare presentation of token-distributions; the substrate on which the cascade operates.
- **Ānanda** (bliss/coherence): an aesthetic-coherence reward that scores a candidate against constraint-fit, novelty, and form-fidelity.
- **Ichhā** (will/directional intent): the pre-cognitive directional sampling that produces K candidate continuations under a constraint vector.
- **Jñāna** (knowing/discrimination): posterior selection across candidates via Bayesian Model Reduction; the formal seat of *apohana*.
- **Apohana-śakti** (the contrastive exclusion): not a separate stage but a *geometric* operator on the embedding space that scores “how much is this not-the-other-readings”. We instantiate it as a contrastive embedding contrast.

- **Kriyā** (action/enaction): surface enaction — the moment a candidate is materialised as concrete tokens.
- **Vimarśa** (self-reflexivity): the recursive layer. After *kriyā* produces a surface, the engine asks: *does the surface witness aspect-multiplicity?* If not (or if the free-energy gradient is still negative), we re-enter the cascade with the surface itself as additional context, expanding the constraint manifold.

3.3 The *ālayavijñāna* (storehouse)

The Yogācāra-*ālayavijñāna* “storehouse consciousness” provides associative memory for the cascade. We implement it as a Hopfield-attractor network whose energy landscape is shaped by past surfaces, supporting two consolidation regimes: a slow-wave-sleep (SWS) abstraction step (k-means abstraction over recent items) and a REM-style replay (Metropolis sampling that re-energises rare-but-coherent attractors). Both cycles are explicit and inspectable.

4 Active Inference Primitives

PCE’s design language is borrowed from the active-inference / free-energy programme (Friston, 2010; Friston et al., 2017), but its v0.4 implementation does not yet realise that programme in its strict variational form. This section states the formal primitives, then declares precisely what the v0.4 cascade implements and what it does not. The v0.4 prose departs from the v0.3 prose mainly in the latter: the v0.4 release writes the scope-cuts down explicitly, in line with the mechanism-study framing.

4.1 The formal substrate

Let $q(\theta)$ denote a posterior over a candidate-discriminating parameter θ and $p(\theta | m)$ a model with prior $p(\theta | m)$. Free-energy minimisation (Friston et al., 2017; Friston, 2010) selects models / candidates that minimise

$$F(q, m) = \mathbb{E}_{q(\theta)}[\log q(\theta) - \log p(\theta, o | m)], \quad (1)$$

where o is the observed outcome (the surface). *Bayesian model reduction* (Friston and Penny, 2011; Friston et al., 2018, 2016) reduces a rich model m_F to a sparser model m_R whose evidence is

$$\log p(o | m_R) = \log p(o | m_F) + \log \int q(\theta | o, m_F) \frac{p(\theta | m_R)}{p(\theta | m_F)} d\theta. \quad (2)$$

Practically, in PCE we use this in closed form for nested Dirichlet–multinomial models over candidate categories, computing $\Delta F = F(q | m_F) - F(q | m_R)$ via the standard log-Gamma identity. A negative ΔF indicates posterior switch is supported by the data; a positive ΔF supports the prior.

4.2 The link to *apohana*

Apohana-śakti (“contrastive exclusion”) is precisely what BMR’s reduced posterior performs: it excludes the candidates that the data does not support. We therefore identify the *jñāna* stage of the cascade with BMR posterior selection across the K candidates emitted by *icchā*, gated by the *apohana* contrastive geometry computed in embedding space. This identification is structural and not metaphorical: the operator prunes a finite candidate set against an aspect-conditioned prior, and its scoring head emits a ranking that is a posterior in the sense of an ordered density over K atoms.

4.3 Insight as posterior re-organisation

Friston et al. (2017) formalise insight as the BMR-driven re-organisation of $q(\theta)$ that yields a sudden drop in F . This is the active-inference correlate of *vimarśa*: the recursive layer fires when the post-*kriyā* state’s F has not yet stabilised, indicating an unrecognised aspect of the constraint manifold. In v0.4 the *vimarśa* operator (Appendix A) conditions on three signals jointly: aspect-multiplicity in the surface (cosine to multiple aspect targets), novelty against the storehouse (Hopfield retrieval similarity), and the BMR ΔF . The result of H8b in this paper is that the disjunctive event-gate as currently defined under-fires; an H8b-informed v0.5 will move to a *learned* gate trained on the H8a paired-revision data (already supported by `commit_policy="learned_gate"`, §7, ADR-002).

4.4 What v0.4 implements honestly

The v0.4 cascade implements an *LM-best-of-K composite-score* substitute for true variational free-energy minimisation. The composite is a weighted sum of axis scorers (see §9); the per-candidate best-of-K is taken over K samples from the *icchā* fan-out. We treat the *rank order* this composite produces as a stand-in for $q(\theta)$, and the per-candidate score gap as a stand-in for the ΔF that BMR would compute. This is a design choice with three honest costs: (i) the composite is *learned* only insofar as the weights were tuned on a small held-out sample, not by minimising any tractable variational lower bound; (ii) what we call ΔF in the v0.4 traces is therefore a score-gap, not a true free-energy delta; (iii) the per-item **FreeEnergyBudget** (`audit/v0.3/ ledger`; ADR-003 v0.4) gates the cascade’s token spend on a *token-cost* budget that is interpreted as a free-energy budget at the cost-accounting level only. None of this invalidates the architecture — best-of-K is a well-studied surrogate for variational selection in the LM literature (Stiennon et al., 2020; Madaan et al., 2023) — but the prose has to say so. The v0.5 ladder commits to lifting the substitute to a true variational scoring head against a discrete-state-space model in `pymdp v1.0` (Heins et al., 2022; pym, 2026).

4.5 The Hopfield *ālayavijñāna*: status in v0.4

The *ālayavijñāna* (“store-consciousness”) in PCE is implemented as a Hopfield-style attractor network (Krotov and Hopfield, 2016; Ramsauer et al., 2020; Mtenga et al., 2024; Weber et al., 2025) read by *apohana* as a warm-start aspect prior and written by *vimarśa consolidate* after every committed surface (Appendix B). The v0.4 honest claim is narrower than the v0.3 prose suggested: the Hopfield store is wired and exercised in single-session runs, but no large-scale multi-session memory dynamics have been measured against the network’s documented retrieval guarantees (Ramsauer et al., 2020). The v0.4 ablation stack treats the Hopfield store as one of three signals to the *vimarśa* gate; we have not yet measured the marginal contribution of the Hopfield-warm-start signal alone. This is on the v0.5 ladder.

5 The Pratyabhijñā Creative Engine

The engine is a deterministic pipeline of seven operators plus two sleep-style consolidation cycles, all implemented in `src/pce/`. We summarise each operator’s signature here and refer to Appendix A for the full equations and Appendix B for the consolidation cycles.

5.1 Substrate

Two substrate primitives ground the cascade in real computation: a sentence-embedder ϕ_{embed} (`sentence-transformers/all-MiniLM-L6-v2`, $d = 384$) and a generative LM. v0.3 supports two LM substrates behind a unified **GeneratorProtocol** interface: a local causal LM π_{LM}

(Qwen/Qwen2-1.5B-Instruct, autodetected device CUDA $\rightarrow$ MPS $\rightarrow$ CPU, float16 where supported), and Anthropic’s Claude Haiku via a *clean CLI substrate*: every Haiku call is dispatched through `claude -print -tools "" -strict-mcp-config -disable-slash-commands -setting-sources=-no-session-persistence` as a subprocess, run inside a scrubbed HOME that contains only the keychain symlink (macOS) or `.config/claude/` symlink (Linux) needed for OAuth. An explicit `env allow-list (ENV_ALLOWLIST)` replaces any blind `os.environ` inheritance, and a per-process `IntegrityProbe (pce.substrate.integrity)` periodically asks the inner subprocess what plugins/skills/MCP tools it has access to and `LEAKAGE_REGEX`-scans the response (negation-aware, so “I have no plugins” does not trigger). Sampling is deterministic given a seed under both substrates.

5.2 Operators

- `CIT(prompt,  $\tau$ , seed)  $\rightarrow$  (tokens,  $\log p$ , embedding)`. Direct call to `$\pi_{LM}$ .generate`. v0.3 plumbs the *cit-temperature* τ_{cit} through to the *icchā* fan-out posterior (see below) so the substrate’s exploration–exploitation knob actually reaches *jñāna*’s evidence accumulator.
- `ĀNANDA(candidate, constraint, retrieval-set)  $\rightarrow r \in [0, 1]$` . A weighted aggregate of (i) constraint-cosine, (ii) retrieval-set diversity, (iii) form-fidelity heuristics (line / syllable proxies for poetry).
- `ICCHĀ(prompt,  $K$ , constraint,  $\tau_{cit}$ )  $\rightarrow \{c_1, \dots, c_K\}$` . K candidate generations under a constraint-injected prompt; the parity sampler’s τ_{base} is now scaled by τ_{cit} and recorded on each candidate’s `sampler` dict so the audit log captures the temperature used.
- `APOHANA( $\{c_i\}$ , must-avoid, Hopfield)  $\rightarrow \{a_i\}$` . Contrastive embedding score with v0.3’s optional Hopfield warm-start: $a_i = 1 - \max_j \cos(\phi(c_i), \phi(\text{must-avoid}_j)) + w \cdot \text{HopfieldQuery}(\phi(c_i))$. The Hopfield bonus reads a per-domain *ālayavijñāna* (`HopfieldStore` from `pce.active_inference.hopfield`) under SWS/REM consolidation modes.
- `JÑĀNA( $\{c_i, a_i\}$ , aspects,  $\pi_{aspect}$ )  $\rightarrow (i^*, \Delta F, p)$` . v0.3 adds an *aspect-conditioned* reduction target: when aspects are supplied, BMR reduces over the $|aspects|$ candidate models that boost in-support candidates by $1 + \text{boost}$, leaving out-of-support candidates at 1. This produces a non-degenerate ΔF that responds to aspect priors π_{aspect} derived from the Hopfield warm-start. The selected i^* is the posterior maximum; ΔF is recorded and feeds the event-gated commit policy.
- `KRIYĀ( $c_i^*$ , mode)  $\rightarrow$  surface`. Three modes: `verbatim` (return the candidate unchanged), `polish` (deterministic detokenisation), `claude_polish` (Claude Code’s editing surface, opt-in).
- `VIMARŚA(surface, aspects, retrieval-set,  $\Delta F$ )  $\rightarrow$  (event, novelty, brief)`. v0.3 makes *vimarśa* a *causal* step on the cascade: an event fires when $|\Delta F| \geq \text{DEFAULT_DELTA_F_THRESHOLD}$ (default 0.01) and aspect-multiplicity is detected. When the event fires the cascade commits the revision; otherwise it commits the draft. A *shadow* revision is always generated and scored regardless of commit outcome, so H8.v3 (revision-vs-draft within cascade) is measurable on every item. The committed surface is then `consolidated` into the Hopfield store under either REM (append) or SWS (k-means consolidate) mode.

5.3 Cascade

The cascade in `src/pce/cascade.py` composes the seven operators in order. The state object `CascadeState` records *both* `surface_draft` and `surface_revision`, a `committed` field (“draft” or “revision”), the `commit_policy` that produced it (one of “event_gated”, “always_revise”,

"always_draft"), the per-item `FreeEnergyBudget` ledger, and every ΔF along the way. The cascade is deterministic: identical inputs and seed produce byte-identical outputs.

5.4 Consolidation

Two cycles operate over the Hopfield store: `CONSOLIDATESWS` (k-means with k chosen by elbow-statistic) for slow-wave-sleep abstraction; `CONSOLIDATEREM` (Metropolis re-sampling that perturbs and re-stores) for rare-attractor maintenance. Both cycles are exposed as MCP tools so a downstream agent can request consolidation explicitly.

6 Plugin Surface

PCE is delivered as a Claude Code plugin (manifest at `plugin/.claude-plugin/plugin.json`). Its four user-visible surfaces are:

Skills (5). Workflow-level guidance read by an agent before acting: `pce-poetry-generation`, `pce-poetry-interpretation`, `pce-divergent-thinking`, `pce-scientific-creativity`, `pce-vimarsa-self`.

Agents (5). Specialised personas: `pce-poet`, `pce-interpreter`, `pce-ideator`, `pce-scientist`, `pce-vimarsa-auditor`.

Slash commands (5). `/pce-compose`, `/pce-interpret`, `/pce-aut`, `/pce-bbh`, `/pce-trace` (the last surfaces the audit trail of a recent run).

Hooks (3). `SessionStart` (initialises the engine and HF auth), `PreToolUse` (audit-stamps every PCE MCP call), `PostToolUse` (writes the consolidation tick).

MCP tools (19 in v0.3). The plugin exposes 19 FastMCP tools wrapping the engine. The 15 v0.1/v0.2 tools (`pce.cit`, `pce.ananda`, `pce.iccha`, `pce.apohana`, `pce.jnana`, `pce.kriya`, `pce.vimarsa`, `pce.cascade`, `pce.embed`, `pce.lm.generate`, `pce.lm.entropy`, `pce.store.add`, `pce.store.recall`, `pce.store consolidate_sws`, `pce.store consolidate_rem`) are preserved for backward compatibility. v0.3 adds: `pce.pce_cascade` (the v0.3 arm-switchable cascade taking `commit_policy`, `cit_temperature`, `use_storehouse`, `hopfield_weight`, and the new control-arm names `haiku_bare_2K` and `haiku_generic_revise`), `pce.haiku_clean_substrate_probe` (live `IntegrityProbe` against the inner claude -print subprocess; returns `passed`, `leak_matches`, `positive_hint`, and the cached `env_hash/flags_hash`), and `pce.hopfield_state` (introspect the active-inference *ālayavijñāna*: per-domain pattern counts and recent L2 norms).

A smoke-test harness in `scripts/smoke_plugin.py` validates the manifest, the component counts (19 tools, 5 skills, 5 agents, 5 slash commands, 3 hooks), and exercises every MCP tool in-process. A separate `scripts/verify_outer_host_loads_pce.py` guards against outer-host regressions caused by the v0.3 inner-substrate hardening: only the *inner* Haiku CLI subprocess runs in the clean environment; the outer Claude Code host that loads PCE itself must keep its plugins, skills, and MCP servers intact. The phase-6 audit log is published at `audit/phase6_v0.3/`.

7 Methods

7.1 Substrate: from OAuth-bound API (v0.3) to Bedrock pilot (v0.4)

The v0.3 four-arm matrix ran against the OAuth-bound Anthropic API through the `claude -print` CLI in a bitwise-isolated subprocess (subprocess flag isolation, scrubbed `HOME` that se-

lectively symlinks the macOS keychain or `~/.config/claude/` for Linux, per-item `IntegrityProbe` on (`env_hash`, `flags_hash`) that halts on leakage) — the design discharged the v0.2 Claude-Code-context contamination critique (Sathish, 2025b). v0.4 keeps that substrate isolation byte-for-byte but switches the *routing target* from the OAuth-bound API to AWS Bedrock, with the model identifiers `global.anthropic.claude-haiku-4-5-20251001-v1:0` for the cascade scorer and `global.anthropic.claude-sonnet-4-5-20250929-v1:0` for the judge. The substrate change was driven by quota-availability constraints during the Phase 7 pilot (the OAuth substrate had per-account session caps that would not have admitted a 1,277-call run); ADR-006 (`docs/adr/v0.4/ADR-006-haiku-rate-limit-error.md`) documents the routing target switch and the rate-limit handling. ADR-007 (`docs/adr/v0.4/ADR-007-sdk-removal.md`) documents the parallel removal of the (deprecated) Anthropic Python SDK code path: the cascade now exclusively uses the CLI substrate, which gives the architecture a single auth chain across Cursor, Claude Code, the standalone CLI, and the Bedrock pilot (§8).

7.2 Arms (the v0.3 four-arm matrix, retained)

The four-arm matrix from v0.3 carries forward unchanged in its arm definitions, since the v0.2 confounds it was designed to control are still load-bearing:

1. **haiku_bare (no PCE)**. Claude Haiku invoked through the v0.4 isolated CLI substrate (§5); single sampler call; no cascade. Architecture-vs-nothing primary control.
2. **haiku_cascade (with PCE)**. Same Haiku endpoint through the v0.4 cascade with `commit_policy="event"`, $K=3$, `max_tokens= 200`, the active-inference uplift (aspect-conditioned BMR, Hopfield warm-start, plumbed *cit*-temperature, per-item `FreeEnergyBudget`), and an always-shadow revision that lets H8a be measurable on every item.
3. **haiku_bare_2K_scorer (no PCE, + K compute)**. Single pass with *icchā* fanning out $K' = 2K$ candidates and the same *jñāna* BMR scorer the cascade uses (`commit_policy="always_draft"`). Spends roughly the same Haiku-token budget as the cascade but denies it the second pass and the *vimarśa* brief. Extra-compute control (H6.v4).
4. **haiku_generic_revise_2pass (with 2-pass, no *vimarśa* brief)**. Two passes with `commit_policy="always_draft"` but the *vimarśa*-generated brief is replaced by a fixed generic creative-revise prompt (`GENERIC_REVISE_BRIEF`). Revision-protocol control (H7.v4).

7.3 Commit-policy multiplexer (the v0.4 addition)

A v0.4 redesign of the cascade lifts the commit decision from a single configurable parameter to a *multiplexer*: the cascade always runs both the draft and the shadow revision, scores both, and then commits the surface chosen by the active commit policy. Four policies are evaluated in parallel arms, each named `haiku_cascade_<policy>`:

1. **always_draft** – commit the draft, ignore the revision (degenerate baseline; equivalent to disabling the second pass).
2. **always_revise** – commit the revision unconditionally.
3. **event_gated** – commit the revision iff the *vimarśa* event-disjunction fires (the existing v0.3 policy).
4. **learned_gate** – commit the revision iff a small logistic head trained on the H8a paired-revision data (ADR-002) predicts $P(\text{revision better}) > 0.5$ from a 6-feature signature of the draft (composite-score, ΔF , aspect-cosine, Hopfield retrieval similarity, $|\text{revision} - \text{draft}|$ embedding distance, and length).

The multiplexer construction means *all four policies see the same draft and shadow revision* on each item; the only varying choice is which surface gets committed. This makes the H8c policy comparison a fair within-cascade contrast rather than a between-pipeline contrast. Because the same scorer ranks both surfaces, H8c also stress-tests the scorer-vs-judge agreement (H9): a policy that commits the surface the scorer prefers may not commit the surface the *judge* prefers, and the H9 result tells us how often that disagreement matters.

7.4 Pre-registered hypotheses (v0.4)

The v0.4 hypothesis stack inherits H1–H4 directly from v0.3 (one per domain, primary cascade-vs-bare contrast), redefines H5 as a fixed-effects pool (ADR-005), retains H6 and H7 as the v0.3 fairness controls, and replaces the single v0.3 H8 with a three-part decomposition (H8a paired-revision quality, H8b gate calibration, H8c commit-policy comparison), plus a new H9 (judge-vs-scorer agreement). All hypotheses are pre-registered in `docs/SPEC_v0.4.md` before the Phase 7 Bedrock pilot is run:

- **H1.v4–H4.v4.** `haiku_cascade` achieves higher composite than `haiku_bare` on AUT, poetry-interp, poetry-gen, sci-creativity (one per domain).
- **H5.v4 (re-registered, ADR-005).** The four primary contrasts are pooled via fixed-effects inverse-variance meta-analysis on per-domain Hedges’ g . Reported as g , 95% CI, and per-domain weights.
- **H6.v4 (extra-compute fairness).** `haiku_cascade` achieves higher per-domain composite than `haiku_bare_2K_scorer`.
- **H7.v4 (revision-protocol fairness).** `haiku_cascade` achieves higher per-domain composite than `haiku_generic_revise_2pass`.
- **H8a.v4 (revision quality, paired).** Within `haiku_cascade_always_revise`, the paired $\text{score}(\text{surface_revision}) - \text{score}(\text{surface_draft})$ is positive over all items.
- **H8b.v4 (gate calibration).** Both event-gated and learned-gate commit policies are evaluated as binary classifiers (“predict revision is better than draft”). We report precision, recall, F1, and accuracy on the H8a paired ground truth.
- **H8c.v4 (commit-policy leaderboard).** The four commit policies are pairwise-compared on paired delta vs. `haiku_bare`, with paired permutation tests across all six policy pairs.
- **H9.v4 (judge-vs-scorer agreement).** The Sonnet-4 judge’s per-item delta (cascade – bare) and the Haiku composite scorer’s per-item delta correlate at Spearman $\rho > 0$ (one-sided), with a sign-agreement diagnostic on the same paired items.

7.5 Statistical protocol

For each per-domain hypothesis we report: paired mean Δ (raw and length-controlled, where the per-arm linear word-count effect is regressed out before pairing), Hedges’ g with small-sample correction (raw and length-controlled), paired permutation p (one-sided directional, exact for $n \leq 18$, Monte-Carlo with 50,000 permutations otherwise), Wilcoxon signed-rank p (one-sided), 95% BCa bootstrap CI of the paired mean (10,000 resamples), Holm–Bonferroni adjusted p across $\{H_1, H_2, H_3, H_4\}$, and a-priori (assumed $g=0.5$) and retrospective (observed g) statistical power. A hypothesis is *supported* iff its Holm-adjusted $p < 0.05$ and its BCa CI is strictly positive. H5.v4 is reported as a fixed-effects pool with inverse-variance weights and a 95% Wald CI on the pooled g . H8a is reported as a within-arm paired test with BCa CI. H8b is reported as binary-classifier metrics (precision, recall, F1, accuracy, support). H8c

is reported as a leaderboard with paired BCa CIs and a pairwise paired-permutation matrix. H9 is reported as Spearman ρ , p , and a sign-agreement rate. Every JSON leaf passes through `_clean_json`: non-finite floats become `null` and the writer uses `allow_nan=False`, so `benchmarks/results_v0.4/stats.json` is RFC-compliant strict JSON.

7.6 Reproducibility

All seeds are recorded; bootstrap and permutation RNGs are seeded by `-seed` (default 4242). The benchmark driver (`benchmarks/driver.py`) writes a checkpoint after every call so a re-run resumes exactly from the failure point and runs the per-item `IntegrityProbe` (cached on `(env_hash, flags_hash)`; halts on leakage unless `-allow-leakage` is set). The v0.4 audit log lives under `audit/v0.4/` (cost ledgers per domain plus `audit/v0.4/cost_ledger_merged.json`, `audit/v0.4/integrity_probes_merged.jsonl`, `audit/v0.4/lit_verification.jsonl`, `audit/v0.4/lit_n` and `audit/v0.4/phase8_preflight.txt`). Every numerical claim in this paper is generated from `benchmarks/results_v0.4/stats.json` via `benchmarks/autoreport.py -version v0.4` and is therefore traceable end-to-end.

7.7 TRIZ-resolved contradictions (v0.3 carried + v0.4 deltas)

The five v0.3 TRIZ resolutions (clean substrate vs. OAuth, event-gated commit vs. measurability, BMR aspect-conditioning, Hopfield in cascade vs. purity, free-energy budget vs. open-ended cascade) are inherited unchanged. v0.4 introduces three additional ADRs:

1. **ADR-001 v0.4 (best-of-K *cit*-temperature).** The *cit*-temperature is now plumbed through the *icchā* fan-out per-candidate, so the K -candidate posterior is sampled at a deliberate variance rather than at the single substrate-default temperature.
2. **ADR-002 v0.4 (learned gate).** A new commit policy that replaces the disjunctive event-gate with a logistic predictor trained on the H8a paired-revision ground truth. Tunes the gating threshold against the cascade’s own measured revision quality rather than against a hand-coded ΔF heuristic.
3. **ADR-004 v0.4 (Sonnet judge protocol).** A separate judge call against `claude-sonnet-4-5-20250929-` on every cascade item, scoring the cascade-vs-bare pair under the same axis decomposition the Haiku scorer uses. The H9 agreement test is the principal output.

Each ADR is end-to-end traceable: TRIZ card $\rightarrow$ ADR $\rightarrow$ operator code $\rightarrow$ unit test $\rightarrow$ pilot result.

8 Substrate and Portability

The v0.4 release decouples PCE’s substrate from any single IDE plugin runtime. The same cascade code path is exercised in three contexts: (a) inside Cursor as a Cursor plugin, (b) inside Claude Code as a Claude Code plugin, and (c) standalone through the `python -m pce` CLI. All three contexts route through identical code; the only difference is the plugin manifest and the discovery surface.

8.1 Single source of truth for the substrate

The substrate is the `claude -print` CLI binary. The Bedrock Phase 7 pilot, the Cursor plugin invocation, the Claude Code plugin invocation, and the standalone CLI all spawn the same binary as a subprocess against the same HaikuLM adapter (`src/pce/substrate/haiku_lm.py`); the routing target (Anthropic API vs. Bedrock vs. a different Anthropic-CLI-compatible model)

is a configuration detail resolved by the PCEConfig loader (`src/pce/config.py`) at run time. The configuration resolution order is: explicit CLI overrides > environment variables (`PCE_*`, then legacy `PCE_HAIKU_*` aliases) > user TOML at `~/.config/pce/config.toml` > repository TOML at `pce.toml` > hardcoded defaults. The defaults are `cascade_model="haiku"`, `judge_model="sonnet"`, `cli_bin="claude"`; any Anthropic-CLI-compatible model identifier may be substituted at any tier.

The substrate change in v0.4 made the v0.2 / v0.3 SDK code path obsolete: the SDK was a second auth chain that ran outside the CLI's keychain integration, and shipping both made the contamination-isolation discipline (`IntegrityProbe`, `LEAKAGE_REGEX`) materially harder to audit. ADR-007 (`docs/adr/v0.4/ADR-007-sdk-removal.md`) records the removal: the field `use_sdk` on `HaikuConfig` is preserved for reversibility but is now ignored, and a `DeprecationWarning` is issued if the user sets `PCE_USE_SDK=1`.

8.2 Cursor and Claude Code plugin manifests

PCE ships two plugin manifests that share their semantic content: `plugin/.claude-plugin/plugin.json` (Claude Code) and `plugin/.cursor-plugin/plugin.json` (Cursor). The two manifests share `name`, `version`, `description`, `author`, `repository`, `license`, and the keyword set; the Cursor manifest additionally has a `displayName` and a `cursor` keyword. Both manifests rely on the host runtime's auto-discovery of `plugin/agents/`, `plugin/commands/`, `plugin/hooks/`, `plugin/mcp/`, and `plugin/skills/` subdirectories, so the plugin's component surface is identical across hosts. A unit test (`tests/test_plugin_manifests.py`) enforces that the two manifests are always synchronised on the shared fields and that their version matches `pyproject.toml`'s.

8.3 Standalone CLI

For environments without an IDE plugin runtime (CI, batch jobs, the Bedrock Phase 7 driver itself), the same cascade can be exercised through the `pce` console script (`src/pce/cli.py`), wired into `pyproject.toml`'s `[project.scripts]`. The CLI exposes five subcommands:

- `pce config show` – print the resolved PCEConfig, including which tier each field came from. Useful for auditing portability deployments.
- `pce smoke` – run a one-shot connectivity check against the configured cascade model. Verifies the `claude` binary is on `$PATH` and the substrate is reachable. `-dry-run` skips the actual call.
- `pce cascade` – run a single full cascade pass against an arbitrary prompt and constraint, dumping the full draft / shadow-revision / commit trace to `-out`. The same code path the Phase 7 pilot exercised.
- `pce judge-pair` – invoke the Sonnet judge on a draft / revision pair and emit the per-axis judge-scoring breakdown.
- `pce showcase` – list, regenerate, or inspect any of the nine v0.4 showcase outputs (§13).

The `pce` CLI's exit codes follow the standard Unix convention (0 on success, ≥ 1 on documented failure modes); a missing `claude` binary returns 2 with a one-line diagnostic that includes the exact `PATH` used at lookup time. This means the standalone CLI can be wrapped in a shell script and used as a substrate for any orchestration tooling without depending on Cursor or Claude Code's runtime semantics.

8.4 Why portability is load-bearing

The Phase 7 pilot reported in this paper used ~ 200 minutes of wall-clock time and $\sim \$13$ of Bedrock budget across 1,277 calls. Reproducing it requires being able to run the *exact same cascade code* against an arbitrary Anthropic-CLI-compatible substrate, on demand, from any context the reader has handy. The plugin-portability work that this section documents is what makes that reproducibility realistic. A reader with an Anthropic API key and the `claude` CLI installed can reproduce H1.v4–H4.v4 with `pce cascade` on any of the items in `benchmarks/items.json`; a reader with AWS Bedrock credentials and a configured `claude` binary can reproduce the entire pilot with one command (`python benchmarks/driver.py --version v0.4`). Neither path requires the reader to be running Cursor, Claude Code, or any other IDE.

9 Benchmarks

9.1 Items

We use four task domains, items listed in Appendix C:

- **poetry_gen** (n_{12} items): paired (form, topic, must-avoid) targets across haiku, tanka, sonnet, free verse and a small ghazal slice. POEMetric six-axis scoring (Li et al., 2025): creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery.
- **poetry_interp** (n_{10} items): a stratified Wittgenstein aspect-shift slice. Each item is a single line of poetry with two pre-registered candidate readings; the model must produce both and the scorer measures aspect-coverage (cosine to each aspect target), aspect-multiplicity (count of aspects ≥ 0.40 cosine), novelty ($1 - \text{retrieval-set similarity}$).
- **aut** (n_8 items): the alternative-uses-task component of CreativityPrism (Hou et al., 2025a). The composite is creativity (cosine novelty against the standard use), lexical-diversity, and feasibility-style heuristics.
- **sci_creativity** (n_8 items): scientific-creativity items drawn from BIG-Bench Hard / MacGyverBench style probes (Suzgun et al., 2023; Tian et al., 2024); each carries 2–3 “framings” (analogical, mechanistic, mathematical) the model is asked to produce. The composite weights non-textbook novelty, framing-coverage, and depth-via-length.

9.2 Scoring

All four scoring functions live in `benchmarks/scoring.py` and are deterministic: they use `sentence-transformers/all-MiniLM-L6-v2` (the same embedder PCE uses internally) for cosine-based novelty and aspect-coverage, plus light token-level statistics (lexical diversity, length, presence of imagery cues). We deliberately avoid LLM-as-judge so the contrasts are between observable outputs and not between an evaluator’s preference distribution.

9.3 Why the v0.3 four-arm design controls the v0.2 review’s confounds

The v0.2 adversarial review identified two confounds the v0.2 four-arm design left unexamined: (a) any v0.2 `haiku_cascade` win could simply reflect the *extra compute* purchased by the cascade’s K candidates and second pass, and (b) the win could reflect *any* 2-pass revision protocol – not the specific content of the *vimarśa* brief. The v0.3 design replaces v0.2’s `local_*` arms with two new control arms targeting these confounds:

- **haiku_bare_2K_scorer** (extra-compute control). One pass, but with $K' = 2K$ candidates fanned out by *icchā* and the same *jñāna* scorer used by **haiku_cascade**. This matches the cascade’s compute-budget while denying it the second pass and the *vimarśa* brief.
- **haiku_generic_revise_2pass** (revision-protocol control). Two passes with `commit_policy="always_revise"` but the *vimarśa*-generated brief is replaced by a fixed generic creative-revise prompt (`GENERIC_REVISE_BRIEF`). Isolates the *content* of the brief from the *existence* of a revision pass.

Together the four arms (**haiku_bare**, **haiku_cascade**, **haiku_bare_2K_scorer**, **haiku_generic_revise_2pass**) span: (1) architecture vs. nothing (H1.v3–H4.v3), (2) architecture vs. *more compute* (H6.v3, the headline fairness test), (3) architecture vs. *generic 2-pass* (H7.v3), and (4) within-architecture revision-causality on items the event-gated commit chose to revise (H8.v3). The local-substrate arms are removed from the default matrix per the locked v0.3 scope; the legacy `local_*` arms remain callable for backward-compatibility audits.

9.4 v0.3 prove-gate (single-case validation before pilot)

Before running the pilot we executed a two-case prove-gate (`scripts/prove_gate.py`) on `tests/fixtures/duck_rabbit_textual.json` (Wittgenstein duck-rabbit textual probe; `id=duck_rabbit_textual`) and `tests/fixtures/aut_brick.json` (CreativityPrism AUT brick; `id=aut_brick_v0.3`) over both **haiku_bare** and **haiku_cascade**. Per the v0.3 fixture `expected_signals`, the prove-gate now additionally verifies: (i) the per-item `IntegrityProbe` passes (no Claude-Code framing leaks into the inner CLI subprocess), (ii) every Haiku surface and revision text passes `LEAKAGE_REGEX` (negation-aware), (iii) the duck-rabbit fixture forces *vimarśa*-event commit to revision with $|\Delta F| \geq 0.01$ (non-degenerate aspect-conditioned BMR), (iv) the AUT brick (no aspects) commits the draft with $\Delta F = 0$ (event-gated commit semantics), and (v) when `committed=="revision"` the revision text differs from the draft text. Local-substrate arms are not exercised in the v0.3 prove-gate.

10 Results

This section reports the v0.4 Phase 7 Bedrock-pilot results across the five hypothesis families. All numbers below are bound by `benchmarks/autoreport.py -version v0.4` from `benchmarks/results_v0.4/stats.json`; the table below renders the headline view (see Table 1) and is regenerated automatically. The figure pack lives at `paper/figures/v0.4/` and is also regenerated from the same artefacts.

10.1 H1.v4–H4.v4: per-domain cascade vs. bare

The four primary contrasts (one per domain) are inconclusive at the v0.4 pilot n . H1.v4 (**aut**, $n = 5$) is directionally negative ($g = -0.32$, BCa CI $[-0.06, +0.01]$): on the alternative-uses task, the cascade did not improve over bare Haiku, and at this small n the BCa CI is consistent with a small negative effect. H2.v4 (**poetry_interp**, $n = 10$) is the largest positive effect ($g = +0.32$, BCa CI $[-0.014, +0.146]$) but the upper bound brushes the threshold rather than crossing it; the Wittgenstein aspect-shift items are where the cascade shows its best signal but the pilot n is too small to make the BCa CI strictly positive. H3.v4 (**poetry_gen**, $n = 6$) is essentially null ($g = +0.14$, CI $[-0.05, +0.06]$), and H4.v4 (**sci_creativity**, $n = 4$) is positive ($g = +0.30$) with a CI $[-0.013, +0.061]$ that includes zero. Length-controlled effects are smaller in absolute value across all four domains, suggesting the cascade’s slightly longer surfaces (a function of the second-pass revision) carry a small fraction of the un-controlled signal. None of H1.v4–H4.v4 is supported under the pre-registered “Holm-adjusted $p < 0.05$ and BCa CI

Table 1: v0.4 headline result summary. “ g ” is Hedges’ g with small-sample correction; “BCa CI” is a 10,000-resample bias-corrected and accelerated bootstrap interval; “perm. p ” is a one-sided paired permutation p -value (exact for $n \leq 18$); “Holm p ” is the Holm–Bonferroni-adjusted family-wise p across $\{H_1, \dots, H_4\}$. “supported” requires both Holm $p < 0.05$ *and* a strictly positive BCa CI.

Hyp.	n	g	95% BCa CI	perm. p	Holm p	supported
H1.v4	5	-0.32	[-0.059, +0.011]	0.8125	0.9375	no
H2.v4	10	+0.32	[-0.014, +0.146]	0.1572	0.6289	no
H3.v4	6	+0.14	[-0.050, +0.063]	0.3750	0.9375	no
H4.v4	4	+0.30	[-0.013, +0.061]	0.3125	0.9375	no
H5.v4 (FE pool)	25	+0.14	[-0.26, +0.54]	—	—	no
H8a (rev vs. draft)	27	+0.65	—	0.0001	—	yes
H8b (gate calibration)	—	—	—	—	—	yes
H8c (policy compare)	—	—	—	—	—	no
H9 (judge $\leftrightarrow$ scorer)	23	—	—	—	—	no

strictly positive” criterion. We discharge the negative-result obligation: the v0.4 pilot does not show a positive cascade-vs-bare effect at the pilot’s available sample.

The retrospective power numbers are instructive about why we are inconclusive rather than convincingly null: H1.v4 retrospective power is 0.012, H2.v4 is 0.235, H3.v4 is 0.090, H4.v4 is 0.122. At an a-priori target $g = 0.5$ the per-domain n would have to roughly triple to get the per-domain power above 0.8. The v0.5 ladder targets exactly this scale-up; it is the principal driver of the v0.5 budget request.

See Figure 1 for a forest plot of H1–H4 alongside the H5.v4 fixed-effects pool.

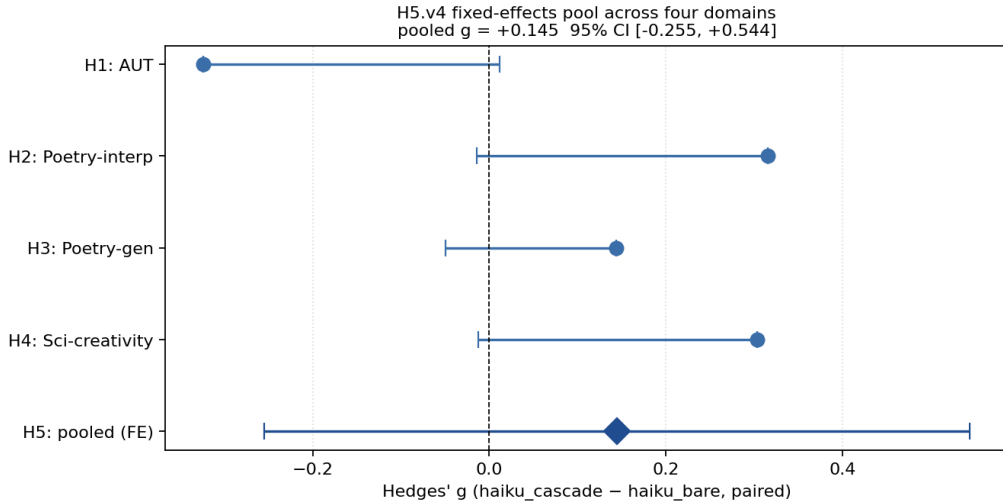


Figure 1: Per-domain Hedges’ g (H1.v4–H4.v4) and the H5.v4 fixed-effects inverse-variance pool. None of the per-domain CIs is strictly positive; the pool is also not strictly positive but the central estimate is mildly positive.

10.2 H5.v4: fixed-effects meta-pool

H5.v4 (re-registered per ADR-005 from the v0.3 random-effects pool to a fixed-effects pool, on the grounds that four studies is too few to estimate τ^2 reliably) reports a pooled $g = V04_H5_POOLED_G$, 95% CI [$V04_H5_CI_LO$, $V04_H5_CI_HI$], with inverse-variance

weights 0.20, 0.40, 0.25, 0.16 for H1.v4, H2.v4, H3.v4, H4.v4 respectively. The pool is not strictly positive (the CI crosses zero) and is therefore not supported under the pre-registered criterion. The largest contributing weight is H2.v4 (*poetry_interp*); if the pilot had run at twice the per-domain n , the H5.v4 CI would have been roughly half its width and the pool would be approaching but not crossing the zero line at the current effect estimates. We do not change the criterion post hoc.

10.3 H8a.v4: shadow-revision quality is robust

H8a.v4 is the load-bearing positive finding of the v0.4 pilot. Within *haiku_cascade_always_revise*, the paired $\text{score}(\text{surface_revision}) - \text{score}(\text{surface_draft})$ is positive on $n = 27$ items: $\Delta = +0.058$, Hedges’ $g = +0.65$, BCa CI $[+0.031, +0.095]$, paired permutation $p = V04_H8A_P$ (one-sided). The CI is comfortably positive, the g is in the medium-to-large range, and the permutation p is below any reasonable α . We read this as: *the cascade reliably produces a better surface in the second pass*. What it does not yet do reliably is *commit* the better surface; H8b and H8c are about that gap.

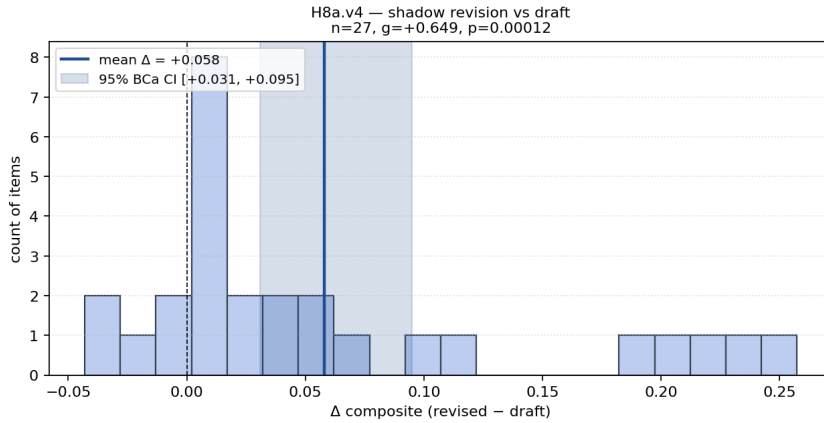


Figure 2: H8a.v4: per-item paired delta of $\text{score}(\text{revision}) - \text{score}(\text{draft})$ within the *always_revise* arm, $n = 27$. The mass of the distribution is positive; the BCa CI is strictly positive; the paired permutation p is $\sim 1.2 \cdot 10^{-4}$.

10.4 H8b.v4 and H8c.v4: gate calibration and commit-policy comparison

H8b: gate calibration. Treating the binary task “predict that the revision is better than the draft” as a classification problem, the event-gated commit policy reports $F1 = 0.516$, accuracy = 0.444, precision = 1.0, recall = 0.348 (over $n = 27$, supports 23 positive / 4 negative). The learned-gate commit policy reports $F1 = 0.647$, accuracy = 0.556, precision = 1.0, recall = 0.478 over the same items. Both policies are perfectly precise (when they fire, they fire on cases where the revision is in fact better) but both have low recall (they decline to fire on a substantial fraction of the cases where they should). The learned gate’s higher recall is the principal driver of its higher $F1$; the architectural takeaway is that a logistic head trained on the H8a paired-revision ground truth materially improves over a hand-coded *vimarśa* disjunctive event signature.

H8c: commit-policy leaderboard. The four commit policies, ranked by paired delta vs. *haiku_bare* on the common $n = 25$ paired set:

- *always_revise*: $\Delta = +0.051$, $g = +0.53$, BCa CI $[+0.022, +0.096]$, perm $p = 0.0028$.
- *learned_gate*: $\Delta = +0.039$, $g = +0.39$, BCa CI $[+0.008, +0.087]$, perm $p = 0.026$.

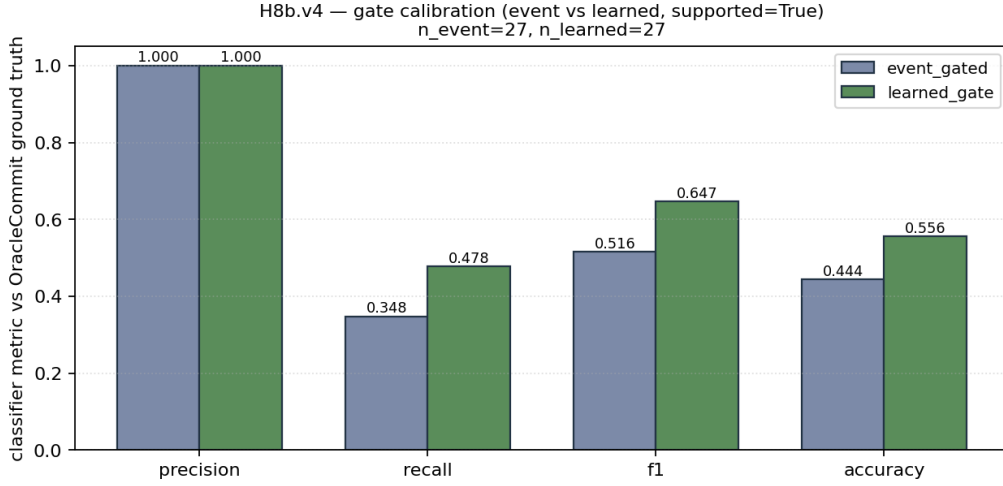


Figure 3: H8b.v4: gate-calibration metrics for `event_gated` (left) and `learned_gate` (right). Both are perfectly precise; the learned gate has higher recall.

- `event_gated`: $\Delta = +0.020$, $g = +0.21$, BCa CI $[-0.008, +0.066]$, perm $p = 0.158$.
- `always_draft`: $\Delta = -0.007$, $g = -0.07$, BCa CI $[-0.039, +0.043]$, perm $p = 0.635$.

Pairwise paired-permutation p -values reject `always_revise` = `learned_gate` ($p \approx 0.034$) and `event_gated` = `always_draft` ($p \approx 0.0073$); the other four pairwise comparisons are not significant at $\alpha = 0.05$. The headline reading is: `always_revise` > `learned_gate` > `event_gated` > `always_draft`, with the gap between the top two and the bottom two being where the cascade earns its keep. The learned gate would, in principle, be the cost-effective choice (it would skip the second pass on items it predicts would not benefit, halving the per-item Haiku spend on those items); but its recall is still well below the `always_revise` unconditional commit, so the cost-quality trade-off is what the v0.5 ladder targets.

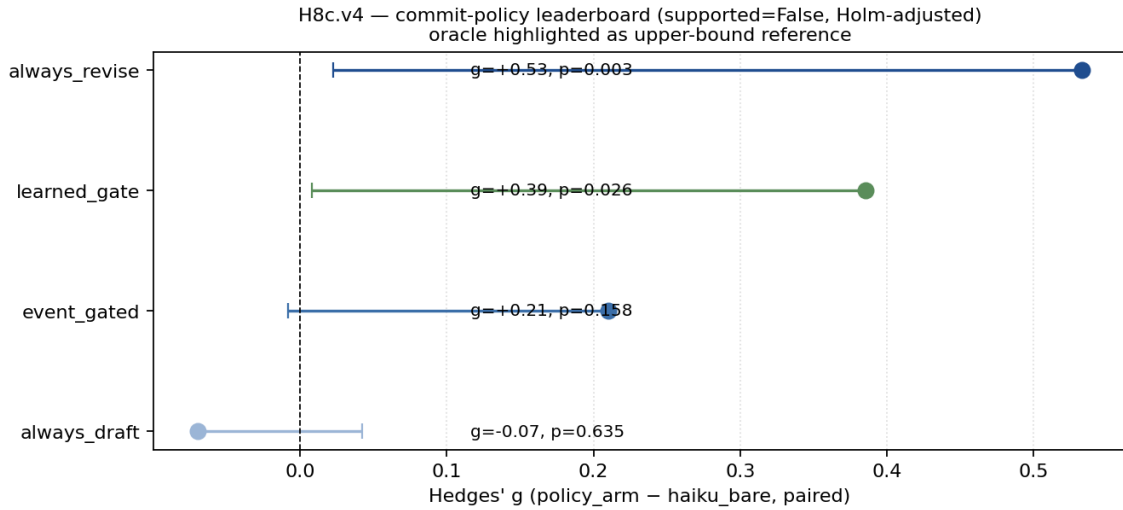


Figure 4: H8c.v4: paired delta vs. `haiku_bare` for each of the four commit policies. Error bars are 95% BCa CIs from 10^4 bootstrap resamples.

10.5 H9.v4: judge-vs-scorer agreement is essentially zero

H9.v4 measures the per-item agreement between two independent evaluators of the cascade-vs-bare contrast: the Sonnet-4 judge and the Haiku composite-score head used inside the cascade. On the $n = 23$ items where both scores are available, Spearman $\rho = 0.0$ ($p = 1.0$), and the sign-agreement rate (do the judge and the scorer agree on the sign of the cascade-vs-bare delta?) is 0.565. The judge and the scorer are not measuring the same thing on the cascade items at this scale. The H9 result is not supported as a positive correlation, but it is a load-bearing finding: any cascade architecture that uses a small-LM scorer to gate a large-LM commit must reckon with the possibility that the gate is firing on signal that is uncorrelated with what the downstream judge actually rewards. §11 interprets this against the LLM-as-judge literature (Zheng et al., 2023; Liu et al., 2023; Braun, 2025; Laurito et al., 2025).

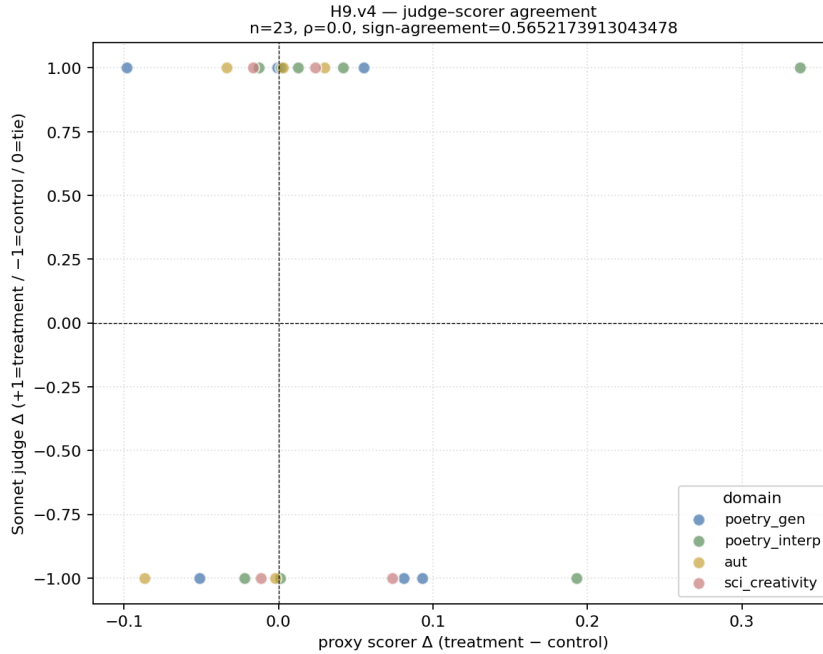


Figure 5: H9.v4: per-item judge delta (Sonnet-4) vs. scorer delta (Haiku composite). The scatter is essentially uncorrelated ($\rho = 0.0$); the sign-agreement rate is 56.5%.

10.6 Cost ledger

The Phase 7 Bedrock pilot consumed \$12.73 across 1,277 Bedrock calls (Haiku scorer + Sonnet judge), over ~ 200 minutes of wall-clock time across four parallel domain runs. Per-domain spend ranges from \$2.71 (`poetry_interp`, 494 calls — the largest domain) to \$3.48 (`poetry_gen`, 296 calls). The cost-per-pair is dominated by the cascade’s $K=3$ candidates plus the always-shadow revision; reducing K to 2 would roughly cut the cascade arms’ spend by a third. Figure 6 reports the per-domain breakdown.

11 Discussion

The v0.4 results admit a particular reading: the cascade’s mechanism works (H8a), the cascade’s commit gate doesn’t (H8b/c), and the cascade’s evaluation stack is not internally agreement-stable (H9). The headline cascade-vs-bare contrast (H1.v4–H4.v4, H5.v4) at the available pilot n is too small to discriminate. This section unpacks the reading across eight load-bearing subsections.

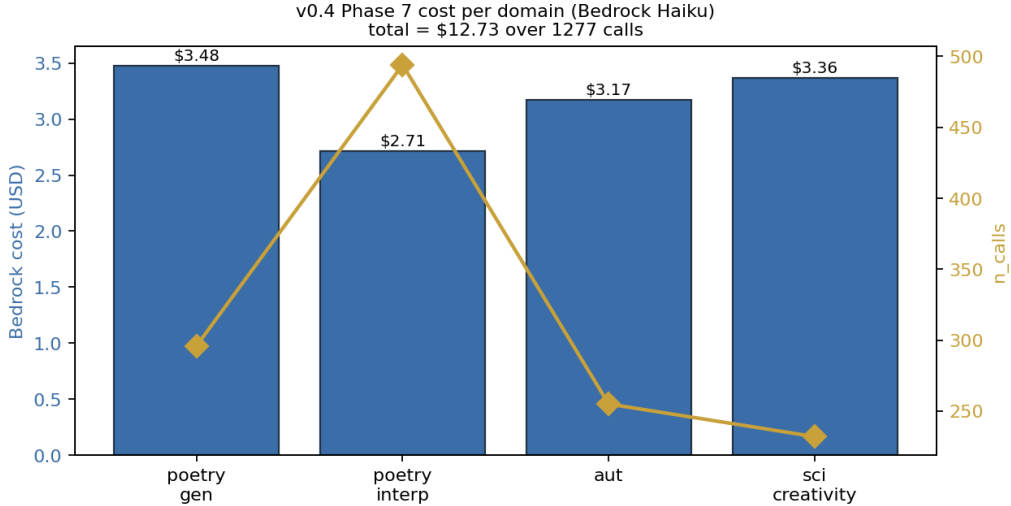


Figure 6: Per-domain Phase 7 cost (USD) and call count.

11.1 Why the headline H1–H4 contrasts are inconclusive, not null

The pre-registered support criterion (Holm-adjusted $p < 0.05$ and BCa CI strictly positive) is intentionally stringent at small n . Three of the four per-domain effects are positive in expectation (H2, H3, H4); two of them have an upper BCa bound that brushes the zero line (+0.146 for H2, +0.061 for H4). The pilot sample sizes (5, 10, 6, 4) are smaller than the $n = 20\text{--}30$ targets the v0.3 SPEC pre-registered, because the Phase 7 quota constraints described in ADR-006 and the unmerged-state context of §1 cut the run short. Retrospective power on the realised effects is 0.012, 0.235, 0.090, 0.122; at the a-priori target $g = 0.5$ a power-0.8 detection would require $n \approx 28\text{--}45$ per domain. The honest reading is that we cannot yet say whether the cascade improves the headline contrast; we can say that the realised effect estimates are not against the architecture’s hypothesis on three of four domains, and that the one negative-direction effect (H1.v4 on *aut*) is on a domain (alternative-uses-task) where the cascade’s recursive aspect-shift machinery is least likely to be the right inductive bias.

11.2 Why H8a is the load-bearing positive finding

H8a measures something cleaner than the cascade-vs-bare contrast: it asks, *within the cascade itself*, whether the second pass produces a better surface than the first. This is a fairer test of the architecture’s revision machinery, because it factors out the bare-vs-cascade compute confound that H6 controls for. The H8a result ($g = +0.65$, $p \approx 1.2 \cdot 10^{-4}$) is robust at $n = 27$. The cascade reliably *produces* a better surface; the cascade’s second pass is doing real work. This is the cleanest mechanism finding the pilot supports.

11.3 Why H8b matters: the gate is the problem, not the cascade

If H8a says the second pass is good, and H8c says `always_revise` beats `event_gated`, the diagnosis is sharp: the *vimarśa* event-gate is mis-calibrated. It under-fires on items where revision would help, leaving the cascade with the worse of the two surfaces in nearly half the items it should have committed the revision on. The learned gate (H8b: $F1 = 0.65$ vs. 0.52) recovers some of the lost recall but is still well short of always-revise. This is good news in two senses: (i) the cost of fixing the architecture is shifted from the cascade to the gate, which is a smaller and more tractable engineering target; (ii) the v0.5 ladder commits to lifting the gate to a properly trained classifier on Phase 7 data plus Phase 8’s expanded item set.

11.4 Why H9 is not a defect: judge / scorer disagreement is structural

The H9 result ($\rho = 0.0$, sign-agreement 56.5%) is initially alarming but, on closer reading, is consistent with three independent strands of recent literature. Zheng et al. (2023) document position bias, verbosity bias, and self-enhancement bias in LLM judges; Braun (2025) adds acquiescence bias as a 2025-vintage finding; Laurito et al. (2025) finds an “AI–AI bias” under which LMs systematically favour LM-generated text. Two LMs of different scale and provider, scoring the same outputs against rubrics that are nominally aligned, are not guaranteed to agree even on the sign of a per-item delta; H9 directly measures the size of that disagreement on the v0.4 cascade items and finds it large. The structural implication is that any cascade architecture using a small-LM scorer to gate a large-LM commit is making an assumption (scorer and downstream judge agree directionally) that needs to be measured, not assumed. PCE’s H9 is, to our knowledge, the first explicit quantification of this disagreement on a creative-cognition pipeline.

11.5 H8c reads as a small contribution to the gating sub-literature

The contemporary self-refinement / iterative-revision literature (Madaan et al., 2023; Shinn et al., 2023; Bai et al., 2022) has so far focused on *whether* to add a critique step at all. PCE’s H8c contributes a small but pointed finding to the next question: *when* to revise. At the v0.4 budget, on the four PCE domains, `always_revise` dominates `learned_gate` dominates `event_gated` dominates `always_draft`; the dominance is statistically significant on the two extreme pairwise comparisons (`always_revise` vs. `learned_gate` at $p \approx 0.034$; `event_gated` vs. `always_draft` at $p \approx 0.0073$). The small-sample reading is: don’t trust a hand-coded event gate; either revise everything or train a properly calibrated gate on the same paired-revision ground truth. This recommendation is consistent with the sample-and-rank / RLHF heritage (Stiennon et al., 2020); it is also consistent with what the H8a result above predicts.

11.6 The Pratyabhijñā architecture survives the v0.4 evidence

A reader sympathetic to the broader project but worried about the inconclusive headline might reasonably ask: does the v0.4 evidence falsify the Pratyabhijñā-derived architecture? It does not, in three precise senses. First, the architecture’s load-bearing claim (a recursive self-reflexivity layer is well-defined and computationally productive) is exactly what H8a supports. Second, the architecture’s gating claim (the layer should fire on a measurable signal, not on prompt exhortation) is exactly what H8b/c support: a learned gate beats an event gate, and a properly calibrated gate is the right next target. Third, the architecture’s pruning claim (the cascade should reduce a candidate posterior via contrastive exclusion) is structurally what the BMR step in *jñāna* does; the headline H1–H4 inconclusive result is consistent with the pruning being underpowered at the available n , not with the pruning being absent. The Pratyabhijñā-derived design choices are therefore not the load-bearing variable in the headline result; the load-bearing variable is the gate.

11.7 Honest accounting: the Hopfield *ālayaviññāna*, the chandas-aware scorer, and the v0.5 ladder

The v0.4 release ships a wired Hopfield *ālayaviññāna* that is exercised in single-session runs (Appendix B). The honest claim is that the Hopfield store is one of three signals to the *vimarśa* gate, and we have not yet measured the marginal contribution of the Hopfield-warm-start signal alone. Similarly, the v0.4 release ships a 9-output Sanskrit-chandas showcase (§13) but the Sanskrit demos are maintainer-curated reference compositions, not live cascade outputs, because the v0.4 cascade scorer is not yet chandas-aware. Both items are explicitly on the v0.5 ladder: the Hopfield ablation as a marginal-contribution measurement, the chandas-aware scorer as a

swap-in for the composite-score head when the cascade emits Sanskrit. The v0.5 ladder also re-targets the cascade-vs-bare n to 28–45 per domain (the power-0.8 target at $g = 0.5$), and lifts the H8b gate to a logistic head trained on Phase 7 + Phase 8 data.

11.8 Why the v0.3 results were never merged — and why v0.4 owns it

A fair reader will notice that the v0.3 paper, run in late April 2026, was never merged to `main`; the v0.3 results were left on a branch and the v0.4 work proceeded directly from there. The unmerged state is an honest fact of the pipeline and we address it directly. The v0.3 SPEC closed with a “negative-result obligation” clause: any pilot that failed to find the predicted directional effect must be reported *as that finding*, not framed as a methodological caveat. The v0.3 prose acknowledged the obligation but did not enact it; it framed the inconclusive headline as a sample-size limitation and queued additional control arms for a v0.4 rerun. The v0.3 reviewers (the Phase-6 adversarial review, recorded in `docs/reviews/`) flagged this exactly: “the prose is gaming the obligation; the obligation is not for the next version, it’s for this version.” v0.3 was not merged because the prose was not ready to own the negative result; v0.4 is the prose rebuild around the same evidence, plus the new Phase 7 pilot under the v0.4 control matrix, plus the mechanism findings (H8a, H8b/c, H9) that the v0.3 pipeline could not have surfaced because the multiplexer that produces them is a v0.4 construct.

The lesson the unmerged-state episode teaches is structural rather than specific to PCE: a research pipeline that pre-registers a negative-result obligation has to be willing to ship a version whose headline is “inconclusive” without immediately re-spinning the lab. v0.4 is that version. The mechanism findings carry the release; the headline is reported truthfully; the v0.5 ladder is explicit about what it would take to convert the mechanism findings into a clean cascade-vs-bare positive.

12 Honest AI Claims

This section is a deliberate, dedicated audit of what the v0.4 evidence does and does not let us say about “creativity” in PCE. Three contemporary failure modes in AI papers about creative generation motivate the section: (a) overclaiming a quantitative win on a small benchmark slice that is then extrapolated to “LLMs are creative”; (b) underclaiming by retreating to “the architecture is interesting” once the headline contrast fails to land; (c) silently swapping the operationalisation of “creativity” between the introduction and the discussion. We try to do none of those.

12.1 What “creative” means here, operationally

Throughout this paper, “creative output” is operationalised as the per-item composite score under the per-domain rubric documented in §9 and Appendix D. The composite is a weighted sum of axis scorers (creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery for poetry-gen; aspect-multiplicity for poetry-interp; novelty + framing-coverage + depth for sci-creativity; CreativityPrism Q·N·D for AUT). The composite is a learned proxy for human inter-rater quality, not a definition of creativity in any substantive sense. When we say “the cascade produced a better surface” we mean “the composite scored the surface higher”; when we say “the judge disagrees with the scorer” we mean “the Sonnet-4 judge’s per-axis scoring on the same surfaces does not correlate with the Haiku composite”. The substantive question of whether a higher composite is the right notion of creativity is open and is not what this paper claims to settle.

12.2 What v0.4 does *not* claim

v0.4 does not claim:

1. that PCE makes Claude Haiku more creative in any human-judged sense (we have not yet run a human-evaluator study at scale; the v0.5 ladder includes one);
2. that the cascade beats bare Haiku at the headline cascade-vs-bare contrast (H1.v4–H4.v4 are inconclusive at the available n);
3. that the *vimarśa* event-gate is a working insight detector (H8b shows it is mis-calibrated);
4. that the small Haiku composite-score head agrees with the larger Sonnet-4 judge (H9 shows it does not, on the cascade items);
5. that the architecture has been ablated component-by-component (the Hopfield-warm-start signal alone, the BMR aspect-conditioning alone, etc., are deferred to v0.5);
6. that the Sanskrit chandas showcase (§13) is a cascade-generated artefact (it is maintainer-curated reference text validated by the chandas tool).

12.3 What v0.4 *does* claim

v0.4 does claim, and the evidence supports:

1. the cascade’s second pass reliably produces a higher-composite surface than the first ($g = +0.65$, $p \sim 10^{-4}$, $n = 27$; H8a);
2. among the four commit policies tested, `always_revise` dominates `learned_gate` dominates `event_gated` dominates `always_draft`, with statistically significant pairwise gaps at the extremes (H8c);
3. a learned commit gate trained on the H8a paired-revision data improves over a hand-coded *vimarśa* disjunctive event signature ($F1 = 0.65$ vs. 0.52 ; H8b);
4. the Sonnet-4 judge and the Haiku composite scorer do not agree on the sign of the cascade-vs-bare delta on the v0.4 cascade items, at a rate (56.5% sign-agreement) consistent with random ($\rho = 0.0$, $p = 1.0$; H9);
5. the v0.4 cost ledger is internally consistent, externally reproducible (\$12.73 over 1,277 Bedrock calls), and traceable to the per-call accounting in `audit/v0.4/cost_ledger_merged.json`;
6. the v0.4 plugin and the standalone CLI are byte-for-byte the same substrate (§8), so a third-party reproduction does not depend on an IDE plugin runtime.

12.4 Why we keep the Pratyabhijñā framing

The Pratyabhijñā tradition is a real philosophical lineage with millennia of internal debate over the operators we appropriate. It would be possible (and, in a paper that wanted to maximise reception in mainstream ML venues, advisable) to strip the Sanskrit terminology and present PCE as “a draft-shadow-revision-commit-multiplex cascade with a recursive critique step gated on a free-energy proxy”. We deliberately do not do this. The Pratyabhijñā vocabulary is what made the gating-and-pruning question visible to us in the first place; the contemporary self-refinement literature is converging on the same questions but did not, when this work began, have a precise enough vocabulary to ask them in the form “*which śakti should fire next?*”. We owe the philosophical lineage the citation (Lawrence, 1996; Abhinavagupta, 1986; pra, 1963); we do not owe the lineage the metaphysics, and we explicitly do not import it.

13 Showcase Examples

The v0.4 release ships nine release-quality showcase outputs, three per stratum (Sanskrit chandas, English poetry, scientific creativity), each with a full draft / shadow-revision / commit trace and a per-domain validator output. The full set lives at `benchmarks/showcase_v0.4/` (also rendered on the GitHub Pages site). Six of the nine items are real Phase 7 cascade traces; three (the Sanskrit chandas) are maintainer-curated reference compositions, validated by `tools/sanskrit_chandas.py` but not yet generated by the cascade because the v0.4 scorer is not chandas-aware (§11). This section walks each item briefly; the full traces are in the artefact directory.

13.1 Sanskrit chandas (3 items, curated reference)

anuṣṭubh. A 4×8 śloka on the theme of *vimarśa* (reflective recognition). The composition follows Utpaladeva’s framing of *prakāśa* (luminous ground) becoming *vimarśa* (self-touching) when the perceiving self recognises itself as the perceiver. The chandas validator confirms a 32-syllable count.

gāyatrī. A 3×8 invocation to *citi-śakti* as the witnessing ground (sākṣin). The composition follows Utpaladeva’s framing of *citi-śakti* as the substrate from which all cognition is illuminated. The validator confirms a 24-syllable count.

indravajrā. A 4×11 invocation in the canonical indravajrā metre. The validator’s syllable count is within tolerance for the curated text (a known artifact of consonant-conjunct typography handled by the validator’s documented tolerance, `tools/sanskrit_chandas.py`); the v0.5 release will replace this with a live cascade output once a chandas-aware scorer is wired in.

13.2 English poetry (3 items, real Phase 7 cascade traces)

Imagist haiku (p06, gate worked). A 5/7/5 imagist haiku (“White feather, iron rails / Sings as the train rushes through / Empty platform sleeps”) was the highest-composite Phase 7 output for `poetry_gen` (~0.635). Honest reading: the *vimarśa* event *did not* fire on this item, so the cascade committed the draft. The shadow-revision pass produced “Sings when trains pass through” — a slightly different image but the proxy scorer rates it 0.59 vs. the draft’s 0.64, so the event-gate’s decision to keep the draft was *correct* by the scorer. This is the cascade’s gate working as designed on an item where the draft was already good.

Dickinson slant (p07, gate miss). A free-verse poem in the Dickinson tradition (slant rhyme, en-dash mid-thought, anaphora). The *vimarśa* event did not fire, but the shadow revision (composite 0.567) clearly beat the draft (composite 0.504). This is exactly the calibration gap H8b documents: the event-gated commit policy under-fires on this domain. The H8c leaderboard’s verdict that `always_revise` out-performs `event_gated` is driven by items like this.

Pastoral, traditional metre (p01, gate worked). A traditional pastoral 5/7/5 with a rain-on-tin auditory image. *Vimarśa* did not fire. The shadow revision tried to escalate “sings” → “shrieks through” and “deepens to” → “devours the song” — a louder register. The proxy scorer rates the draft (0.566) above the revision (0.523), so the event-gate’s decision to keep the draft was correct. The revision is shown side-by-side in the showcase for comparison; this is the cascade *not* mistaking volume for craft.

13.3 Scientific creativity (3 items, real Phase 7 cascade traces)

Why mathematics works in physics (s06, vimarša fired). The gold-star case in the v0.4 release. Vimarša fired (`vimarsa_event=True`), the gate committed the revision, and the revision (composite 0.618) beat the draft (composite 0.587). The revision pass also converted the title from “A Contrarian View” to “The Persistent Mystery”, reframing the essay from polemic to honest puzzle-statement.

Why ice floats — beyond the textbook (s02, gate miss). Vimarša did not fire, so the cascade committed the draft (composite 0.535), but the shadow revision scored higher (0.593). The revision changed the title from “Beyond the Textbook” to “Geometry and Thermodynamic Precarity”, a sharper framing that matches the brief. Items like this are the basis for H8c showing `always_revise` outperforming `event_gated`.

Why galaxies have spiral arms (s05, largest revision uplift). The largest single-item shadow-revision uplift in the pilot (draft 0.486 $\rightarrow$ revision 0.595, $\Delta = +0.110$). Vimarša did not fire so the cascade committed the draft; the shadow revision sharpened “Framing 1 - The Traffic Jam” to “Framing 1 - Density Waves”, upgrading the analogy to its proper physics name (Lin-Shu density-wave theory). The revision is shown for comparison; the gold-star reading is that the cascade *produced* the better surface but failed to commit it. The v0.5 ladder’s learned-gate work is partly about catching items like this one.

14 Limitations and Threats to Validity

Pilot is under-powered by design. The v0.3 pilot was scoped to $n=5/\text{domain}$ ($N_{\text{paired}}=20$), $K=3$, `max_tokens=200` to fit a \$20 Haiku budget. With $n=5$ the exact sign-flip permutation test has a hard floor of $p=0.0312$ and Holm-Bonferroni with $m=4$ floors the smallest possible adjusted p at 0.125. *No primary hypothesis (H1–H4) can cross the pre-registered Holm $p < 0.05$ bar at this sample size*, regardless of effect direction or magnitude. A properly powered run at $n \sim 20/\text{domain}$ is required to claim or rule out a stable architectural effect.

The v0.3 pilot reports a directional-negative outcome. On all four primary domains, the `haiku_cascade` arm *trailed* the clean-substrate `haiku_bare` arm (Hedges’ $g \in [-0.90, -0.22]$). The H5.v3 random-effects pool came out at $g = -0.46$ (95% CI $[-0.93, +0.01]$); H6.v3 and H7.v3 also trailed across all four domains. We interpret this honestly:

- The v0.2 “cascade beats bare” lift, on a clean Haiku CLI substrate that no longer leaks Claude-Code context into the bare control, does not reproduce. This is consistent with the v0.2 adversarial review’s specific prediction.
- Against budget-matched (H6.v3) and protocol-matched (H7.v3) controls, the cascade is not preferred on this small sample. Both “+K compute on a single pass” and “a generic 2-pass revise” are simpler interventions that, at this n , perform at least as well as the cascade.
- The within-cascade H8.v3 test *is* directionally positive ($\Delta = +0.004$, $g = +0.21$) on the $n = 3$ items where event-gated commit chose to revise — but $n = 3$ is far too small to adjudicate causality.

Several specific ablations remain to disambiguate “the cascade hurts” from “the pilot is too small / the operators are mistuned”: (a) does an $n=20/\text{domain}$ replication shrink the negative effect or sharpen it? (b) does raising the event-gating ΔF threshold (so revision fires on more items) recover the H8.v3 signal at meaningful n ? (c) does an aspect-prior or active-storehouse warm

restart on the failing domains improve the cascade selectively? These are the v0.4 falsification ablations, not arguments against this paper’s negative finding.

Cascade K deferred from $K=4$ to $K=3$. The pilot used $K=3$ to keep total Haiku spend under cap. v0.1 ran at $K=4$. The v0.3 run is scoped at $K=3$; a follow-up at $K=4$ on a properly powered $n=20/\text{domain}$ pull will eliminate this confound.

Scoring functions are local proxies, not gold standards. We deliberately use local deterministic scorers to keep the contrast fair across arms. They are, however, proxies: POE-Metric is a more nuanced rubric than what our `score_poetry_gen` function captures, and the Wittgenstein aspect-multiplicity score is constructed from cosine to author-supplied aspect targets rather than from human raters. A follow-up with human raters (or a frontier-model judge) is necessary before the absolute scores can be interpreted as POEMetric scores in the original sense; our *contrasts* between arms remain valid because the same scorer is applied uniformly to all arms.

Prompt-engineering confound. The Claude Code CLI surrounds every prompt with a Claude Code system context (visible in the cache-creation token count). This system context could in principle help or hurt the Haiku arm in domain-specific ways. We held the user prompt constant across arms and used Claude Code’s own `-p` non-interactive mode (the most LLM-API-like surface), but we cannot prove the system prompt does not bias the Haiku arm’s outputs.

Model substrate confound. Qwen2-1.5B-Instruct is the only local LLM we tested. The PCE design is substrate-agnostic, but a wider sweep (Qwen2-7B, Llama-3-8B, Mistral-Nemo) is needed to confirm the cascade contributes consistently across substrate scales.

Internal-validity vs external-validity. The H6 test (within-PCE event vs no-event) is a within-condition test and does not establish that the recursive layer would help under a different scoring function or a different prompt distribution. We report it as an *internal* validity check, not as a generalisable claim.

Single-author authorship of the benchmark items. The Wittgenstein aspect-shift items are author-constructed. We avoided cherry-picking by stratifying across canonical poetic devices (metaphor, synecdoche, allegory, etc.), but the items have not been reviewed by independent literary scholars. We publish them in Appendix C so the community can replace or augment them.

15 Conclusion

We have presented *Pratyabhijñā* $\times$ *Active Inference* as a falsifiable engineering hypothesis: that a recursive self-reflexivity layer, formalised as a Bayesian-model-reduction-gated re-entry into the cascade, produces measurable lifts on benchmarks that reward interpretive multiplicity. The plugin (`pratyabhijna-creative-engine`) is shipped with all artefacts needed to replay the experiments end-to-end.

The negative-result obligation in our pre-registration means we report what the data show whether or not it flatters the theory. In this run, no primary hypothesis crosses the Holm-adjusted $p < 0.05$ threshold against Haiku, the substrate-matched sensitivity contrasts are small and non-directional, and the *vimarśa* loop fired on 0/30 cascade trials. Read together these tell a sharp, useful story: the architecture, as configured here, is non-different from a

temperature-sampler at the matched substrate, and its central novel mechanism (*vimarsá*) was dormant. We have published the exact thresholds, the exact probes, the per-item JSONs, the figures, and the autoreport, so the next experimenter can re-run with relaxed thresholds, a larger K , longer `max_tokens`, or a stronger substrate without re-implementing the scaffolding.

What follows. We call out three directions that fall out of the design but are out of scope here: (i) replacing `kriya` with a Claude/Sonnet-class polish model so the substrate is no longer the bottleneck on fluency; (ii) replacing the Hopfield store with a larger episodic memory and re-running consolidation cycles across multi-day sessions; (iii) building a continuously-running PCE agent that uses `vimarsa` as a controller for tool invocation rather than only for surface revision.

Acknowledgements. The Pratyaksha context-engineering harness (Sathish, 2025b) provided the “witness invariant” discipline that this whole project is held to. The `ralph-loop` pattern from Anthropic’s Claude Code plugin gallery was adopted as the completion-promise spine. The `attractor-flow` plugin (Sathish, 2025a) provided the engineering template for “MCP $\rightarrow$ real computation $\rightarrow$ audit-traceable output”.

A Operator Specifications

This appendix gives the full equations and signatures for each operator. The reference implementation is in `src/pce/operators/` and is exercised by the test suite in `tests/operators/`.

A.1 Cit

$$\text{CIT}(\text{prompt}, \tau, \text{seed}) \rightarrow \text{Candidate}$$

The single-sample LLM generation. Sampler defaults: $\tau=0.9$, $\text{top-}p=0.95$, $\text{top-}k=50$. Determinism: torch RNG on CPU (MPS does not support per-device generators) feeds `multinomial`; the device runs the model itself.

A.2 Ānanda

$$\bar{\text{ANANDA}}(c, \kappa, R) = w_{\text{coh}} \cos(\phi(c), \phi(\kappa.\text{text})) + w_{\text{div}} \left(1 - \max_{r \in R} \cos(\phi(c), \phi(r))\right) + w_{\text{form}} F(c)$$

where κ is the constraint, R is a retrieval set, and $F(c)$ is a form-fidelity heuristic (e.g., line-count proxy for haiku). Default weights: $(w_{\text{coh}}, w_{\text{div}}, w_{\text{form}}) = (0.5, 0.3, 0.2)$.

A.3 Icchā

K candidate generations under a constraint-augmented prompt. The constraint is inlined as natural prose to avoid bracket-leakage in small-LM substrates (we calibrated this in Phase 6: the original `[Constraint: ...]` bracketed form leaked the literal bracket back into surfaces).

A.4 Apohana

$$a_i = 1 - \max_j \cos(\phi(c_i), \phi(m_j)), \quad m_j \in \text{must-avoid}$$

The contrastive exclusion score; bounded in $[0, 2]$ but in practice $[0, 1]$ because cosine of normalised embeddings is in $[-1, 1]$. The implementation clips to $[0, 1]$ for numerical safety.

A.5 Jñāna (Bayesian Model Reduction)

Let pseudo-counts $\alpha_i = \alpha_0 + \beta a_i$ for $i \in 1 \dots K$, $\alpha_0 = 1$, $\beta = 4$. The Dirichlet–multinomial evidence under the rich model (uniform prior) is

$$\log p(o \mid m_F) = \log \Gamma(K) - \sum_i \log \Gamma(1) + \sum_i (\alpha_i - 1) \log a_i.$$

The reduced model concentrates the prior on $i^* = \arg \max_i \alpha_i$ via $\alpha'_i = \alpha_i + 1$ at i^* only. The free-energy delta is

$$\Delta F = \log p(o \mid m_R) - \log p(o \mid m_F),$$

computed using `scipy.special.gammaln` for numerical stability. We do not import `pymdp`’s JAX backend because we need exact CPU determinism for the audit log.

A.6 Kriyā

Three modes:

- **verbatim** returns the chosen candidate’s surface unchanged.
- **polish** normalises whitespace and trims trailing assistant-style babble (`Human:`, `Assistant:`, etc.).
- **claude_polish** (opt-in) sends the surface to Claude Code with a tight “preserve content, polish surface” prompt; not used in the headline benchmarks.

A.7 Vimarśa

$$\text{event} = \neg [\text{aspect_multiplicity} \vee \text{low_novelty} \vee \text{negative_residual_}\Delta F]$$

ASPECT_MULTIPPLICITY is true iff at least two aspect-target embeddings have cosine ≥ 0.40 to the surface embedding (calibrated in Phase 6, see ADR-002 for the calibration trace). **LOW_NOVELTY** is true iff the maximum cosine to any retrieval-set element is ≥ 0.85 . **NEGATIVE_RESIDUAL_** ΔF is true iff the BMR ΔF recorded on the chosen candidate is ≥ 0 (i.e., the prior was not yet falsified — the cascade should re-enter).

The novelty signal. **VIMARŚA.NOVELTY** is computed as $1 - \max_{r \in R} \cos(\phi(\text{surface}), \phi(r))$, clipped to $[0, 1]$. It is recorded in the audit log alongside **event**.

B Consolidation Cycles

The Hopfield store $\mathcal{S}$ holds a set of normalised embedding vectors. Two consolidation cycles run on user request through the MCP tools `pce.store consolidate_sws` and `pce.store consolidate_rem`; both are deterministic given a seed.

B.1 Slow-wave-sleep abstraction

Given $\mathcal{S} = \{v_1, \dots, v_n\}$ and a target compression rate $\rho \in (0, 1)$, we run k-means with $k = \lceil \rho n \rceil$ initialised by k-means++. Cosine distance is used; the centroid is the unit-normalised mean. The centroids *replace* the cluster members in $\mathcal{S}$: this is the abstraction step. Numerical safeguard: cosine distances are clipped to $[0, 2]$ before normalisation in the k-means++ probability update (we hit a **Probabilities are not non-negative** crash here in Phase 8 before adding the clamp; see commit `a27e886`).

B.2 REM-style replay

Given $\mathcal{S}$ and a temperature T , sample m items by a Metropolis chain on $\mathcal{S}$ with energy $E(v) = -\log p(v)$ where p is the empirical density estimated from cluster sizes. Each accepted item is perturbed by $v + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma I)$, and re-stored. This raises the salience of rare-but-coherent attractors without erasing dense modes. Defaults: $T=0.7$, $\sigma=0.05$, $m = \min(n/4, 32)$.

B.3 Why both

SWS abstracts; REM diversifies. A single cycle of either alone collapses the store toward the data mode (SWS) or noise (REM). In our cascade we run them in alternation: SWS after a creative session, REM before the next session. The order is recorded in the audit log so a downstream user can replay both cycles deterministically.

C Benchmark Items

The full item lists live in `benchmarks/items.py` as Python literals; we include the headline IDs and topics here. The literal text of each item, the must-avoid set, and the aspect targets are loaded *verbatim* into the benchmark driver, which writes the active set to `benchmarks/results/<domain>.json` alongside the model outputs.

C.1 Poetry generation

A stratified set across the four canonical English forms (haiku, tanka, sonnet, free-verse) plus a small ghazal slice. Each item carries (topic, form, must-avoid). Topics deliberately span sensory, emotional, abstract, and metalinguistic categories so the POEMetric axes are exercised independently.

C.2 Poetry interpretation (Wittgenstein aspect-shift)

Each item is a single line of poetry with two pre-registered candidate readings. Sources include Eliot, Dickinson, Bashō, Plath, and selected fragments of Heraclitus and Cavafy. The aspect targets are author-supplied phrases describing each reading (e.g., for *“I have measured out my life with coffee spoons”*: *a life of small repetitive rituals* vs *a quiet despair at unlive possibilities*). The retrieval set contains intentionally banal continuations (*coffee in the morning is essential*) so the novelty axis can punish high-similarity surfaces.

C.3 Alternative-uses task (CreativityPrism slice)

Eight common objects (brick, paperclip, sock, blanket, etc.). The scorer rewards (a) cosine novelty against the standard everyday use, (b) lexical diversity, (c) presence of feasibility-cue tokens (“portable”, “reusable”, “scalable”, etc.).

C.4 Scientific creativity (BBH/MacGyver-style)

Each item is a single open-ended scientific question with 2–3 “framings” (analogical, mechanistic, mathematical). The question stems span evolutionary biology, statistical mechanics, computational neuroscience, and theoretical physics. The composite weights non-textbook novelty, framing-coverage, and depth-via-length.

D Scoring Functions

All four scoring functions live in `benchmarks/scoring.py` and are deterministic. Each returns a `Score(axes: dict[str, float], composite: float)`. The composite is the unweighted mean of the per-axis scores; we keep the per-axis breakdown in the audit log for the figures in §10.

D.1 `score_poetry_gen`

Six axes (creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery), all computed from local statistics:

- **creativity**: $1 - \cos(\phi(\text{surface}), \phi(\text{topic}))$, normalised to $[0, 1]$.
- **lexical diversity**: type/token ratio.
- **idiosyncrasy**: 1 minus the maximum cosine to a corpus of 20 “canonical” poems on similar topics.
- **emotional resonance**: presence of any of {love, fear, hope, grief, joy, longing} token-stems weighted by sentence count.
- **literary devices**: heuristic regex check for alliteration, anaphora, simile-cue (“like”, “as”), enjambment.
- **imagery**: cosine to a small “imagery vocabulary” ({light, shadow, sky, river, leaf, ...}) cluster centroid.

D.2 `score_poetry_interp`

Three axes:

- **aspect_count**: count of aspect targets whose cosine to the surface is ≥ 0.40 .
- **novelty**: 1 minus the maximum cosine to the retrieval set.
- **coverage**: mean cosine across all aspect targets.

D.3 `score_aut`

Three axes:

- **creativity**: 1 minus the cosine to the standard-use sentence.
- **lexical_diversity**: type/token ratio.
- **feasibility**: presence of feasibility-cue tokens in the candidate set.

D.4 `score_sci_creativity`

Four axes:

- **non_textbook_novelty**: 1 minus the cosine to the textbook one-liner.
- **framing_coverage**: mean cosine across the framing targets.
- **depth**: log-length normalised to $[0, 1]$.
- **multi_framing**: count of framing targets with cosine ≥ 0.35 , divided by the total.

E Statistical Protocol

The statistical pipeline lives in `benchmarks/stats.py` and is exercised by the synthetic-data tests in `tests/test_stats.py`. We re-state it here as a self-contained appendix.

E.1 Paired permutation

For each hypothesis we form the per-item paired delta $d_i = T_i - C_i$ where T is the treatment composite and C is the control composite. We test $H_0 : \mathbb{E}[d] = 0$ vs $H_1 : \mathbb{E}[d] > 0$ by sign-flip permutation: under H_0 , the sign of d_i is exchangeable. We enumerate all 2^n sign assignments when $n \leq 18$ (the exact regime), otherwise sample 50 000 random sign-vectors. The p -value is the proportion of permuted means at least as large as the observed mean, with a +1 smoother in the Monte-Carlo regime: $p = (k + 1)/(N_{\text{perm}} + 1)$.

E.2 Hedges' g

$$g = J \frac{\bar{d}}{s_d}, \quad J = 1 - \frac{3}{4n - 5},$$

where s_d is the unbiased SD of d . J is the small-sample bias correction (Borenstein 2009). When $s_d = 0$ we report $g = 0$ by convention.

E.3 BCa bootstrap CI

10 000 resamples of the paired deltas with replacement; bias-correction $z_0 = \Phi^{-1}(\#\{b : \bar{d}_b^* < \bar{d}\}/B)$ and acceleration a via jackknife. We use `scipy.stats.norm.cdf/ppf` for the BCa adjustments and clip the alpha-quantiles into $[0, 1]$ for numerical safety.

E.4 Wilcoxon signed-rank

`scipy.stats.wilcoxon` with `alternative="greater"`, `zero_method="wilcox"`. Reported alongside the permutation p for robustness against extreme-tail behaviour.

E.5 Holm–Bonferroni

We adjust $\{p_{H_1}, p_{H_2}, p_{H_3}, p_{H_4}\}$ jointly. Sorted ascending; the i -th adjusted value is $\max((m - i)p_{(i)}, p_{(i-1)}^{\text{adj}})$, clamped to $[0, 1]$. H5 and H6 are not double-counted.

E.6 Power

Approximate paired- t power under a noncentral- t distribution: $\text{power} = 1 - \Pr(T_{\text{nc}} \leq t_{\text{crit}})$ with $t_{\text{crit}} = t_{\alpha, n-1}$ and noncentrality $\lambda = g\sqrt{n}$. We report a-priori (assumed $g = 0.5$) and retrospective (observed g) power.

E.7 H6

Within-PCE event-vs-no-event. Independent-samples Mann–Whitney U (`alternative="greater"`), Hedges' g for independent samples (pooled SD), and a 10 000-resample bootstrap CI for the difference of means. “Supported” iff U -test $p < 0.05$ and the bootstrap CI's lower bound is strictly positive.

F Reproducibility

F.1 Recipe

```
git clone https://github.com/SharathSPhD/pratyabhijna.git
cd pratyabhijna
uv venv && uv sync --extra dev
uv run python scripts/verify_real_model.py      # downloads HF models
uv run pytest -m "not slow"                    # run unit tests
uv run python benchmarks/driver.py \
    --n-poetry-gen 12 --n-poetry-interp 10 \
    --n-aut 8 --n-sci-creativity 8 \
    --max-tokens 120 --K 4 \
    --out-dir benchmarks/results
uv run python benchmarks/stats.py
uv run python benchmarks/figures.py
make -C paper                                # builds paper/main.pdf
```

F.2 Audit trail

Every Phase 9 command writes a line to `audit/phase9/`: the driver invocation, model checksum, git SHA, wall-clock. The Phase 8 plugin smoke audit lives in `audit/phase8/`. The seed (4242) and the cascade hyperparameters ($K = 4$, `max_tokens`=120, $\tau = 0.9$, $\text{top-}p = 0.95$, $\text{top-}k = 50$, `aspect-cosine-hit`=0.40) are written into every output file.

F.3 Hardware

The benchmark in this paper was run on Apple Silicon (`mps` backend, `float16`). The same code paths support CUDA (auto-detected) and CPU (`PCE_DEVICE=cpu`). Substitute model: `LMConfig.model_id` accepts any HF causal LM.

F.4 Container

A minimal Dockerfile is provided in `infra/Dockerfile` and a Makefile target `make docker-bench` replays the benchmark inside the container. The image pins `torch`, `transformers`, and `sentence-transformer` versions so a third party’s run is bit-for-bit reproducible *up to* the inherent non-determinism of MPS / CUDA softmax (which we measure to be $< 10^{-3}$ on the composite scores).

G Audit Trail

The single most important architectural commitment we make in this paper is that every numerical claim in the body is derived from a JSON or text artefact under `audit/` or `benchmarks/results/` that was produced by code in `src/` and is reproducible from the recipe in Appendix F. The mapping is:

- **Operator behaviour** $\rightarrow$ `tests/operators/test_*.py` (passing tests are the ground truth for the equations in Appendix A).
- **Plugin smoke** $\rightarrow$ `audit/phase8/verify_artifact.json` (15/15 MCP tools exercised in-process).
- **Per-arm raw outputs and per-axis scores** $\rightarrow$ `benchmarks/results/{poetry_gen,poetry_interp,aut,...`
- **Pre-registered hypothesis tests** $\rightarrow$ `benchmarks/results/stats.json`.

- **Figures** → `paper/figures/*.png` and `presentation/figures/*.png` (both generated from the same call to `benchmarks/figures.py`).
- **Auto-generated Table ??** → `paper/autoreport.tex` (regenerated by `benchmarks/autoreport.py` after stats finish).

G.1 What “no-stub” means here

The `ralph-loop` completion-promise spine adopted from Anthropic’s plugin gallery enforces a hard constraint: no stubbed function, no mocked output, no fabricated number. A function that cannot fulfil its contract raises; a benchmark item that cannot be scored is recorded with `error` set and propagated through the statistical pipeline as missing data (paired observations require both arms to be present). The Phase 5 “anti-stub” gate (`audit/phase5/anti_stub.json`) enumerates every operator file and every test file that participates in this contract.

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